An Integrated Interframe Stabilization and Fast Imaging Method for Video Synthetic Aperture Radar

Shuo Wang, Guanyong Wang, Yanping Wang, Rui Zhou, Meng Zhao, and Yuxiao Wang

***Abstract*— Video image stability and imaging efficiency are two main problems in the application of current video synthetic aperture radar (ViSAR) systems. High-resolution millimeter-wave ViSAR imaging is susceptible to interframe drift caused by motion errors, which seriously affects video stability. Image-domain-based registration algorithms would be affected by speckle noise, which causes distortions in synthetic aperture radar (SAR) images. Moreover, the high aperture overlapped rate in ViSAR results in redundant computations in multiframe imaging. Therefore, how to achieve multiframe fast imaging with high stability is a significant research direction of ViSAR. In this article, a novel integrated method of fast multiframe imaging and stabilization for video SAR is proposed. This approach mainly contains two aspects. First, in order to address video interframe drift compensation, the generating mechanism of drift error is analyzed from the imaging aspect. A gradient function is constructed to accurately estimate motion error by maximizing the correlation between subaperture images. Next, based on accurate frame motion error estimation, a subaperture complex-weighted fast backprojection (SCFBP) algorithm is proposed, which combines the interframe motion error compensation in the signal domain. Simulations and real measured data experiments illustrate that the proposed method effectively improves the interframe stability and accelerates multiframe imaging for video SAR.**

***Index Terms*— Image stabilization, motion error compensation, subaperture complex-weighted fast backprojection (SCFBP), video synthetic aperture radar (ViSAR).**

I. Introduction

VIDEO synthetic aperture radar (ViSAR) has the advantage of high flexibility in both temporal and spatial dimensions. Compared with conventional synthetic aperture radar (SAR), it provides high-frame-rate image information in the time domain [1], [2]. High-frequency bands, such as millimeter waves, are currently used in ViSAR to enable continuous observation of the interested region and precise tracking of moving targets [3], [4], [5], [6], [7]. However, millimeter-wave ViSAR is sensitive to motion errors during echo data acquisition due to platform vibration and atmospheric turbulence [8], [9]. Motion errors will introduce phase errors during the SAR imaging process, leading to image defocusing and drifts in multiframe images [10], [11], which seriously degrades the tracking and detection performance of moving targets. In addition, the high dynamic of video SAR puts forward higher requirements for real-time imaging. Therefore, how to improve the imaging efficiency while maintaining high interframe stability is a problem that needs to be studied.

In video SAR systems, interframe image stabilization is a critical step to ensure consistency and stability between consecutive frames in the video sequence. In recent years, extensive studies have been conducted [12], [13], [14], [15]. Current mainstream algorithms first acquire multiframe ViSAR images and then apply feature-based methods to register images. The scale-invariant feature transform (SIFT) is the classical feature-based algorithm for image matching. It has been widely used owing to its efficient performance and invariance to scale, rotation, and illumination changes [16], [17], [18]. Several improved SIFT algorithms have been proposed in the literature [19], [20], [21]. To address the issue that the SIFT algorithm does not perform well in matching SAR images affected by strong multiplicative noise, researchers proposed an SIFT-like algorithm specifically for SAR images (SAR-SIFT) and conducted optimizations [14], [15], [22], [23]. However, these methods primarily focus on feature matching in the image domain. Their registration accuracy is highly dependent on the scene, and image distortion may occur during the registration of nonuniform scenes [24], [25]. The imaging mechanism of SAR is inherently distinct from that of optical images, leading to differences in geometric distortion between images. Consequently, methods such as SIFT are not well-suited for SAR images. Moreover, video SAR requires multiframe image registration, whereas the SIFT algorithm is limited to pairwise registration between a reference and a sensed image [26], thereby making interframe stabilization in video SAR time-consuming.

Due to the platform speed limitations, improving the aperture overlap rate of video SAR is necessary to achieve a high frame rate. However, it increases the computational burden

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Shuo Wang is with the School of Electrical and Control Engineering, North China University of Technology, Beijing 100144, China.

Guanyong Wang, Yanping Wang, and Meng Zhao are with the School of Artificial Intelligence and Computer Science, North China University of Technology, Beijing 100144, China (e-mail: guanbingwang@126.com).

Rui Zhou is with Beijing Institute of Radio Measurement, Beijing 100854, China.

Yuxiao Wang is with the Radar Technology Research Institute, School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China.

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because the overlapping parts between consecutive frames are processed repeatedly. To meet real-time demand, it is essential to accelerate multiframe imaging in video SAR. The backprojection (BP) algorithm, as a classical time-domain imaging method, is commonly used in ViSAR imaging owing to its adaptability to different radar geometries [3]. The traditional BP algorithm is time-consuming, which seriously limits the practical application of video SAR [27]. Yegulalp [28] proposed the fast backprojection (FBP) algorithm, which divides the full aperture into subapertures. Low-resolution imaging in cross-range direction is performed on each subaperture separately. When combining the subaperture images into a final high-resolution image, the process involves upsampling the images to full resolution in the cross-range. Song and Yu [29] divided the SAR scene into the general region (GR) and the region of interest (ROI). By using different aperture lengths in different regions, the processing complexity in the ViSAR system is reduced. While these methods improve the computational efficiency of the BP algorithm, they still treat each video frame as an independent unit.

To exploit the overlap between consecutive frames in video SAR to reduce redundant computations. Hawley and Garber [30] and Wahl et al. [31] sequentially form subaperture images at the final desired pixel spacing and store these images. Full-resolution images of different frames can be obtained through coherent summation. This strategy can reduce the computational load of multiframe imaging at the expense of additional storage space. Moses and Ash [32] proposed a recursive backprojection (RBP) algorithm for ViSAR. This algorithm updates SAR images using a low-order autoregression without calculating the pulses used in the past image formation. Low-rank tensor recovery is much more effective compared with compressed sensing (CS) in exploiting the internal correlation of multidimensional data [33], [34]. Pu et al. [35] exploited the redundancy property of multiframe video SAR data, which can be modeled as a low-rank tensor and formulated the video SAR imaging process as a low-rank tensor recovery problem. The research proposed an alternating minimization method to address this problem, which reduces the number of data samples and computational cost.

From the above analysis, it is evident that existing fast imaging methods do not achieve an integrated processing flow of imaging and stabilization. High-frame-rate video SAR still faces the challenge of low efficiency in generating stable video imagery.

This article analyzes the causes of multiframe image drift in video SAR from the perspective of imaging mechanism. Interframe drift errors are modeled and estimated, and an integrated processing framework is proposed that combines multiframe fast imaging with interframe stabilization. This framework achieves image stabilization in the signal domain through interframe drift compensation, thereby enhancing the consistency of video SAR image sequences. Furthermore, it avoids redundant computations arising from overlapping regions between frames and omits the image-domain registration step during multiframe imaging, thus significantly improving computational efficiency. The main innovations are summarized as follows.

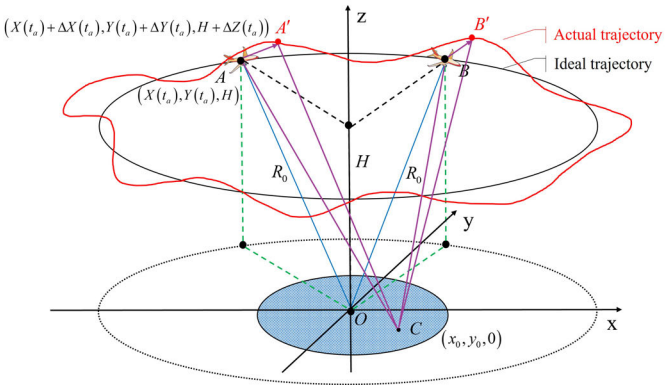


Fig. 1. Geometry of airborne SAR with motion errors.

1) To address the issue of multiframe image drift, the generation mechanism of drift error is analyzed, and an interframe motion error estimation method is proposed by error modeling. A gradient function is constructed to accurately estimate motion error by maximizing the correlation between subaperture images.
2) An integrated framework for efficient and stable imaging is proposed. The framework combines motion error estimation parameters and uses subaperture complex-weighted FBP (SCFBP) for multiframe imaging. Multiple weighted images are generated for each subaperture, which can be combined to approximate the drift compensation function required by any frame, thereby achieving interframe image stabilization while avoiding repeated computation.

This article is organized as follows. Section II presents the motion error model causing interframe image drift and interframe phase error estimation method. Section III introduces the proposed SCFBP imaging algorithm. The flowchart of the integrated fast imaging and stabilization algorithm is summarized in Section IV. Section V details the multiframe point target imaging simulation. Section VI illustrates the processing results of the real data from the airborne ViSAR system and gives the comparison with other algorithms. Section VII concludes this article.

II. Interframe Drift Analysis and Error Estimation

A. Motion Error Model

The SAR imaging geometry is shown in Fig. 1. The black line shows the radar's ideal trajectory, while the red line represents its actual trajectory. In the normal case, the platform moves along the ideal trajectory at a constant speed V_s and altitude H. However, atmospheric turbulence can cause the actual flight path of the aircraft to deviate from its ideal trajectory, resulting in position errors of the antenna phase center (APC), as shown by the purple arrow in the figure.

Let the ideal sensor position be $(X(t_a), Y(t_a), H)$, and a target C in the scene is located at $(x_0, y_0, 0)$. t_a represents the azimuth slow time. Then, the expression for the ideal slant range $R_t(t_a)$ is given by

$$R_t(t_a) = \sqrt{(X(t_a) - x_0)^2 + (Y(t_a) - y_0)^2 + H^2}. \quad (1)$$

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The instantaneous motion error vector is expressed as $(\Delta X(t_a), \Delta Y(t_a), \Delta Z(t_a))$. Taking the motion error into account, the actual slant range from the radar to the target can be written as $R_a(t_a)$ (2), as shown at the bottom of the next page.

Thus, the range error $R_e(t_a)$ caused by the APC deviation is (3), as shown at the bottom of the next page.

According to (3), the received signal of point target with motion error is expressed as

$$s(t_r, t_a) = A_0 \cdot w_r\left(t_r - \frac{2R_t(t_a)}{c} - \frac{2R_e(t_a)}{c}\right) \cdot w_a(t_a - t_{\text{ac}}) \cdot \exp\left(-j4\pi \frac{(R_t(t_a) + R_e(t_a))}{\lambda}\right) \cdot \exp\left(j\pi K_r\left(t_r - \frac{2(R_t(t_a) + R_e(t_a))}{c}\right)^2\right) \tag{4}$$

where A_0 is the amplitude, t_r is the rang time, t_a is the azimuth time, $w_r(\cdot)$ is the range envelope, $w_a(\cdot)$ is the azimuth envelope, λ is the wavelength of radar, c is the light speed, t_{ac} represents the Doppler centroid time, and K_r represents the range frequency modulation (FM) rate.

After range matched filtering to (4), the echo signal becomes

$$s_o(t_r, t_a) = A_0 \cdot \sin c\left(t_r - \frac{R_t(t_a)}{c}\right) \cdot w_a(t_a - t_{\text{ac}}) \cdot \exp\left(-j4\pi \frac{(R_t(t_a) + R_e(t_a))}{\lambda}\right). \tag{5}$$

The Taylor expansion of (3) at $t_a = 0$ is performed, and the high-order terms are negligible based on the boundary condition, which is derived in detail in Appendix A. Let ζ_c, ζ_o, and ζ_q represent the constant, linear-order, and quadratic-order coefficient of azimuth time t_a, respectively. The approximated slant range error expression $\hat{R}_e(t_a)$ is given by

$$\begin{aligned}\hat{R}_e(t_a) &\approx \frac{R_e(0)}{0!} + \frac{R_e'(0)}{1!} \cdot (t_a - 0) + \frac{R_e''(0)}{2!} \cdot (t_a - 0)^2 \\ &= \zeta_c + \zeta_o \cdot t_a + \zeta_q \cdot t_a^2 \\ &\begin{cases} \zeta_c = A_e^{\frac{1}{2}} - B_e^{\frac{1}{2}}; \\ \zeta_o = A_e^{-\frac{1}{2}} \cdot C_e + B_e^{-\frac{1}{2}} \cdot D_e; \\ \zeta_q = \frac{1}{2} \cdot \left(-A_e^{-\frac{3}{2}} \cdot C_e^2 + A_e^{-\frac{1}{2}} \cdot E_e + B_e^{-\frac{3}{2}} \cdot D_e^2 + B_e^{-\frac{1}{2}} \cdot F_e\right). \end{cases}\end{aligned} \tag{6}$$

The expressions from A_e to F_e are presented in detail in Appendix B.

B. Video SAR Interframe Drift Analysis

In video SAR imaging, a high frame rate is ensured by dividing the full-aperture data into subapertures at regular azimuth intervals and processed using an overlapping aperture approach. Subaperture division reduces the data volume for each imaging operation, thereby enhancing real-time performance.

As shown in Fig. 2, let the total number of subapertures in the video SAR echo data be M and the total azimuth time of multiple frames be T_a. Let N represent the number of subapertures in each frame, and the azimuth time corresponding to each subaperture is T_s

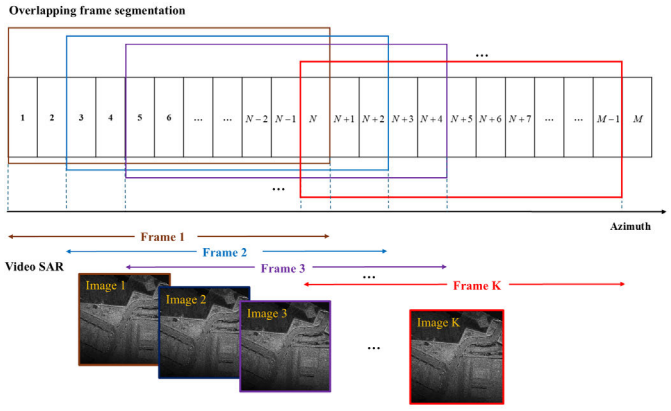


Fig. 2. Schematic of overlapping aperture division in ViSAR imaging.

$$T_s = T_a / M. \tag{7}$$

The azimuth duration of any frame is

$$T_f = N \cdot T_s. \tag{8}$$

Denote N_2 as the number of nonoverlapping subapertures between frames. The azimuth time for the Kth frame can be expressed as

$$t_K = t_a' + (K-1) \cdot N_2 \cdot T_s; t_a' \in [0, T_f] \tag{9}$$

where t_a' represents the azimuth time of the first frame.

From (5) and (6), the echo data $s_K'(t_r, t_K)$ for the Kth frame are obtained

$$\begin{aligned} &s_K'(t_r, t_K) \\ &= s'(t_r, t_K) \cdot \exp\left(-j\frac{4\pi}{\lambda} \cdot \left(\zeta_c + \zeta_o \cdot (t_K) + \zeta_q \cdot (t_K)^2\right)\right) \\ &= s'(t_r, t_K) \cdot \exp\left(-j\frac{4\pi}{\lambda} \cdot \zeta_c\right) \\ &\quad \cdot \exp\left(-j\frac{4\pi}{\lambda} \cdot \zeta_o \cdot \left(t_a' + (K-1) \cdot N_2 \cdot T_s\right)\right) \\ &\quad \cdot \exp\left(-j\frac{4\pi}{\lambda} \cdot \zeta_q \cdot \left(t_a' + (K-1) \cdot N_2 \cdot T_s\right)^2\right) \end{aligned} \tag{10}$$

where $s'(\cdot)$ denotes the ideal signal expression, as given in the following:

$$s'(t_r, t_a) = A_0 \cdot \sin c\left(t_r - \frac{R_t(t_a)}{c}\right) \cdot w_a(t_a - t_{\text{ac}}) \cdot \exp\left(-j4\pi \frac{R_t(t_a)}{\lambda}\right). \tag{11}$$

By simplifying (10), the following expression can be obtained:

$$s_K'(t_r, t_K) = s'(t_r, t_K) \cdot \Theta_K \cdot \exp\left(-j\frac{4\pi}{\lambda} \cdot \zeta_o \cdot t_a'\right) \cdot \exp\left(-j\frac{4\pi}{\lambda} \cdot \zeta_q \cdot \left(t_a'\right)^2\right)$$

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$$\cdot \exp\left(-j\frac{4\pi}{\lambda}\cdot((2K-2)\cdot N_2\cdot T_s)\cdot\zeta_q\cdot t_a'\right). \tag{12}$$

The constant phase term Θ_K can be expressed as follows:

$$\Theta_K = \exp\left(-j\frac{4\pi}{\lambda}\cdot\zeta_c\right)\cdot\exp\left(-j\frac{4\pi}{\lambda}\cdot\zeta_o\cdot((K-1)\cdot N_2\cdot T_s)\right) \cdot\exp\left(-j\frac{4\pi}{\lambda}\cdot\zeta_q\cdot((K-1)\cdot N_2\cdot T_s)^2\right). \tag{13}$$

The quadratic phase term in (12) causes defocusing of the main lobe, which severely degrades the quality and resolution of SAR images. In conventional single-frame SAR imaging, autofocus algorithms are employed to estimate and compensate for this term. The defocusing phase term can be expressed as follows:

$$H_{\text{defocus}} = \exp\left(-j\frac{4\pi}{\lambda}\cdot\zeta_q\cdot\left(t_a'\right)^2\right). \tag{14}$$

However, video SAR typically requires multiframe imaging. During this process, the linear phase term induced by the quadratic coefficient causes the offset of point targets in each frame, and the offset accumulates as the number of frames increases. Consequently, the drift phenomenon occurs in continuous image sequences, which is not considered in previous single-frame imaging. The expression of the drift phase term is

$$H_{\text{drift}} = \exp\left(-j\frac{4\pi}{\lambda}\cdot((2K-2)\cdot N_2\cdot T_s)\cdot\zeta_q\cdot t_a'\right). \tag{15}$$

C. Interframe Phase Error Estimation Method

To achieve the interframe image stabilization in video SAR, it is necessary to estimate the quadratic coefficient and compensate for the phase term in (15). During multiframe imaging, the quadratic error coefficient of each frame is estimated using an autofocus method. However, differences in motion compensation (MOCO) between frames introduce residual phase errors with slow variation. Interframe phase discontinuity will affect the consistency of multiframe imaging.

To ensure the continuity of interframe motion error compensation, this article constructs a full-aperture motion error compensation function corresponding to multiple frames by estimating the phase error. Given the substantial volume of echo data collected by video SAR, a practical approach is to extract a subset of the data for the rapid estimation of the quadratic coefficient of the slant range error.

Assuming that $\hat{N}$ pulses are extracted from the original echo for error estimation, the signal can be expressed as

$$\hat{s}_o\left(t_r,\hat{t}_a\right) = s_o\left(t_r,t_a\left(\tilde{n}\right)\right),\tilde{n}\in\left[\hat{n}_1,\hat{n}_2\right] \qquad \hat{N} = \hat{n}_2-\hat{n}_1+1 \tag{16}$$

where $\hat{n}_1$ and $\hat{n}_2$ are the first and last pulse sequence numbers of the extracted echo, respectively.

The imaging space is divided into grids composed of $x_n \times y_n$ pixels, and the extracted echo is imaged by the BP algorithm. Suppose that the APC at slow time $\hat{t}_a(\hat{n})$ is located at $P(X(\hat{t}_a(\hat{n})), Y(\hat{t}_a(\hat{n})), H), \hat{n}\in[1,\hat{N}]$. The slant range $R_{ij,\hat{n}}$ from radar to the pixel $P_{ij}(x_i, y_j, 0)$ is

$$R_{ij,\hat{n}} = 2\|P-P_{ij}\|_2 = 2\sqrt{\left(X\left(\hat{t}_a\left(\hat{n}\right)\right)-x_i\right)^2+\left(Y\left(\hat{t}_a\left(\hat{n}\right)\right)-y_j\right)^2+H^2} \tag{17}$$

where $\|\cdot\|_2$ denotes the $L2$-norm, $i\in(0,x_n]$, $j\in(0,y_n]$.

The position index $\text{ID}_{ij,\hat{n}}$ of pixel P_{ij} in the echo at the azimuth time $\hat{t}_a(\hat{n})$ is

$$\text{ID}_{ij,\hat{n}} = \text{round}\left[\left(\frac{R_{ij,\hat{n}}-2R_{\min}}{c}\right)\cdot Fs\cdot\text{ratio}+1\right] \tag{18}$$

where $R_{\min}$ represents the shortest slant range, Fs is the sampling rate, ratio is the range oversampling factor, and round$(\cdot)$ denotes the rounding operation.

To estimate the quadratic coefficient ζ_q, the slant range error term can be expressed as

$$\ddot{R}_{\hat{n}} = \zeta_q\cdot\left(\hat{t}_a\left(\hat{n}\right)\right)^2. \tag{19}$$

For the extracted echo, if the final image is obtained by summing two subaperture images IM1 and IM2, the subimages can be expressed as follows:

$$\begin{aligned}\text{IM1} &= \sum_{i=1}^{x_n}\sum_{j=1}^{y_n}\sum_{\hat{n}=1}^{\hat{N}/2}\hat{s}_o\left(\text{ID}_{ij,\hat{n}},\hat{n}\right)\cdot\exp\left(j2\pi\frac{R_{ij,\hat{n}}}{\lambda}\right)\cdot\exp\left(j\frac{4\pi}{\lambda}\cdot\ddot{R}_{\hat{n}}\right)\\ &= \sum_{i=1}^{x_n}\sum_{j=1}^{y_n}\hat{I}_1(i,j)\end{aligned} \tag{20}$$

$$\text{IM2} = \sum_{i=1}^{x_n}\sum_{j=1}^{y_n}\sum_{\hat{n}=\hat{N}/2+1}^{\hat{N}}\hat{s}_o\left(\text{ID}_{ij,\hat{n}},\hat{n}\right)\cdot\exp\left(j2\pi\frac{R_{ij,\hat{n}}}{\lambda}\right)\cdot\exp\left(j\frac{4\pi}{\lambda}\cdot\ddot{R}_{\hat{n}}\right)$$

$$R_a(t_a) = \sqrt{(X(t_a)+\Delta X(t_a)-x_0)^2+(Y(t_a)+\Delta Y(t_a)-y_0)^2+(H+\Delta Z(t_a))^2} \tag{2}$$

$$R_e(t_a) = R_a(t_a)-R_t(t_a) = \sqrt{(X(t_a)+\Delta X(t_a)-x_0)^2+(Y(t_a)+\Delta Y(t_a)-y_0)^2+(H+\Delta Z(t_a))^2} - \sqrt{(X(t_a)-x_0)^2+(Y(t_a)-y_0)^2+H^2}. \tag{3}$$

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$$= \sum_{i=1}^{x_n} \sum_{j=1}^{y_n} \hat{I}_2(i, j) \tag{21}$$

where $\exp\{j2\pi R_{ij,\hat{n}}/\lambda\}$ is used to compensate for the Doppler frequency shift of pixel P_{ij} at different azimuth times [36]. $\hat{I}_1(i, j)$ and $\hat{I}_2(i, j)$ represent the value of the focused pixel at image position (i, j) for IM1 and IM2, respectively.

Similar to the methods presented in [37] and [38], the procedures of motion error estimation and compensation are iteratively performed. Each hypothesis yields a set of subimages. To evaluate whether the image stabilization is satisfactory, the similarity between (20) and (21) is calculated using image correlation.

The object function F can be calculated as follows (22), as shown at the bottom of the next page.

The average values of the subimages are

$$m1 = \frac{\text{IM1}}{x_n \cdot y_n}; m2 = \frac{\text{IM2}}{x_n \cdot y_n}. \tag{23}$$

If the normalized correlation sum is close to unity, the output of object function F in (22) is reduced, and the drift between subapertures is suppressed. If not, another quadratic coefficient must be assessed. The gradient descent method can be used to determine the optimal solution of the quadratic coefficient under the constraint of minimizing the object function value [37], [38], [39], [40].

Notably, the gradient descent algorithm easily converges toward local correlation maximum and produces the suboptimal solution due to the improper initial condition, which degrades the estimated precision and convergence speed [40]. To address this issue, the initial error search set is established, and a penalty term can be introduced to limit the search direction. By selecting the solution with the minimum objective function value as the global optimal solution, the problem of local convergence can be effectively mitigated.

III. Subaperture Complex-Weighted Fast BP Algorithm

To improve image quality and obtain a stable video stream, the traditional BP algorithm applies interframe MOCO and sliding windowing in each frame separately [29]. Due to the high interframe overlap rate of video SAR, the subaperture images are repeatedly calculated during the processing of different frames. Hawley and Garber [30] proposed a fast imaging method based on subaperture weighted for sidelobe suppression by using the overlapping characteristics of adjacent frame data. However, this method does not account for interframe motion error compensation, and thus, the problem of multiframe image drift remains unresolved.

To address the above issue, an SCFBP algorithm is proposed in this article. The flow of the proposed algorithm is shown in Fig. 3, which mainly includes three steps.

Step(a): For any given subaperture, during the BP imaging process, the pulse contributions are weighted differently. Three separate output images are generated concurrently by using the constant, linear, and quadratic pulse weights a, b, and c with respect to azimuth time t_a.

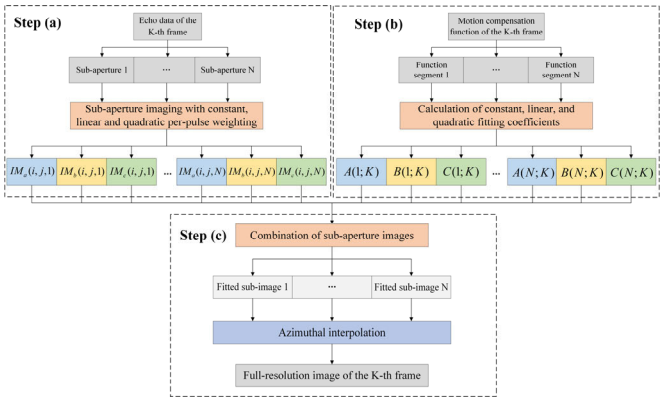


Fig. 3. Flow of SCFBP algorithm.

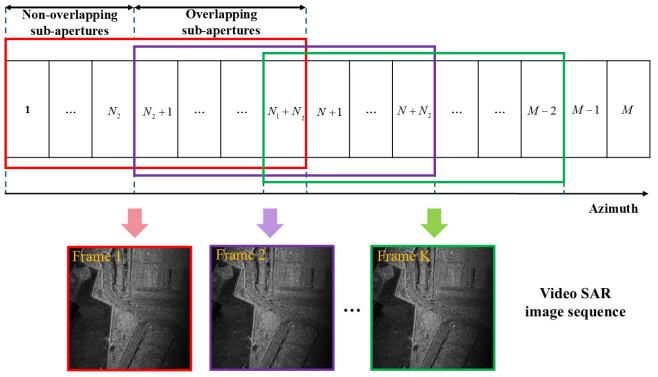


Fig. 4. Diagram of overlapped aperture processing.

Step(b): According to the first and last positions of the current frame pulse, the corresponding segment is extracted from the overall motion error compensation function and combined with windowing to form the required MOCO window function. The constant, linear, and quadratic complex fitting coefficients related to each subaperture image are then computed.

Step (c): This step involves the weighted summation of the subaperture images. The complex fitting coefficients are assigned to the subaperture images with different per-pulse weights, enabling these images to be combined to produce any quadratic approximation to the desired MOCO window function within the subaperture. Subsequently, azimuth interpolation is performed on each subaperture image individually, and the resulting images are summed to obtain the full-resolution image for the current frame.

A. Subaperture Imaging With Per-Pulse Weighting

Video SAR processes echo data by dividing the full synthetic aperture time into multiple subapertures [28]. This approach enables the repeated use of subaperture data to meet the requirements of continuous imaging and high frame rate. For a continuous monitoring scene with cyclic acquisition, the final image frames are usually equal in length in the azimuth direction.

The diagram of overlapping frame segmentation is illustrated in Fig. 4. Assume that the echo data of multiframe video SAR consist of T azimuth pulses, which are divided into M subapertures with equal interval. Let N denote the number of subaperture images in each frame and N_1 denote the

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number of overlapping subapertures between adjacent frames. The number of nonoverlapping subapertures, denoted as N_2, is then given by $N_2 = N - N_1$. Accordingly, the frame number N_f can be calculated as

$$N_f = \frac{M - N}{N_2} + 1. \tag{24}$$

Let n_s denote the number of pulses contained in each subaperture, which is given by $n_s = T/M$. The azimuth time t_{as} of the mth subaperture can be expressed as

$$t_{\text{as}} = t_a(\eta), \eta \in [1 + (m-1) \cdot n_s, \min(m \cdot n_s, T)], m \in M. \tag{25}$$

According to (5) and (25), the echo data s_{os} corresponding to the mth subaperture are

$$s_{os} = s_o(t_r, t_{\text{as}}). \tag{26}$$

Similar to the derivation in Section II, the slant range $R_{ij,t_{\text{as}}}$ from radar position P' to the pixel P_{ij} at azimuth time t_{as} is

$$\begin{aligned} R_{ij,t_{\text{as}}} &= 2\|P' - P_{ij}\|_2 \\ &= 2\sqrt{(X(t_{\text{as}}) - x_i)^2 + \left(Y(t_{\text{as}}) - y_j\right)^2 + H^2}. \end{aligned} \tag{27}$$

The position index $\text{ID}_{ij,t_{\text{as}}}$ of pixel P_{ij} at this azimuth time is

$$\text{ID}_{ij,t_{\text{as}}} = \text{round}\left[\left(\frac{R_{ij,t_{\text{as}}} - 2R_{\min}}{c}\right) \cdot Fs \cdot \text{ratio} + 1\right]. \tag{28}$$

The MOCO window function of each frame is divided into multiple subfunction segments at the same interval as the subapertures. For each subaperture, the subfunction segments can be fit in the form of $y = C \cdot x^2 + B \cdot x + A \cdot 1$, where $1, x$, and x^2 serve as the basis functions, and A, B, and C are the undetermined fitting coefficients.

During the BP imaging process for any given subaperture, three output images are generated with per-pulse constant, linear, and quadratic weighting. The role of these weighted images is to provide the basis functions required for quadratic polynomial fitting.

It should be noted that these weighted images do not inherently determine the specific shape of the fit curve. They need to be assigned fitting coefficients and summed to approximate the motion-compensation window function.

Let a, b, and c represent the per-pulse weights applied during the imaging process to generate three independent images. For multiframe video SAR data, if the number of azimuth pulses in each aperture between frames remains consistent, the same per-pulse weights can be used to generate images for any subaperture in any frame. In this case, the calculation formula of a, b, and c can be expressed as

$$a = \left(\frac{n_{\text{re}} - 1}{n_s - 1}\right)^0, a \in [0, 1], n_{\text{re}} \in [1, n_s] \tag{29}$$

$$b = \left(\frac{n_{\text{re}} - 1}{n_s - 1}\right)^1, b \in [0, 1], n_{\text{re}} \in [1, n_s] \tag{30}$$

$$c = \left(\frac{n_{\text{re}} - 1}{n_s - 1}\right)^2, c \in [0, 1], n_{\text{re}} \in [1, n_s] \tag{31}$$

where n_{re} represents the relative value of the azimuth pulse sequence in the subaperture. The detailed derivations of (29)–(31) are presented in Appendix C.

At the azimuth time t_{as} of the mth subaperture, the Doppler phase term $\phi_{ij,t_{\text{as}}}$ of the pixel P_{ij} is

$$\phi_{ij,t_{\text{as}}} = \exp\left\{j2\pi \frac{R_{ij,t_{\text{as}}}}{\lambda}\right\} \tag{32}$$

where λ is the wavelength of carrier frequency.

After phase compensation for each backprojected pulse reflected by the pixel point P_{ij}, the weights a, b, and c are assigned. By applying this approach, each pixel in the imaging grid is sequentially traversed. Finally, three subimages IM_a, IM_b, and IM_c with different weightings of the mth subaperture can be obtained

$$\text{IM}_a(i, j, m) = \sum_{t_{\text{as}}=t_a(\eta_1)}^{t_a(\eta_2)} s_{\text{os}}\left(\text{ID}_{ij,t_{\text{as}}}, t_{\text{as}}\right) \cdot \phi_{ij,t_{\text{as}}} \cdot a, m \in M \tag{33}$$

$$\text{IM}_b(i, j, m) = \sum_{t_{\text{as}}=t_a(\eta_1)}^{t_a(\eta_2)} s_{\text{os}}\left(\text{ID}_{ij,t_{\text{as}}}, t_{\text{as}}\right) \cdot \phi_{ij,t_{\text{as}}} \cdot b, m \in M \tag{34}$$

$$\text{IM}_c(i, j, m) = \sum_{t_{\text{as}}=t_a(\eta_1)}^{t_a(\eta_2)} s_{\text{os}}\left(\text{ID}_{ij,t_{\text{as}}}, t_{\text{as}}\right) \cdot \phi_{ij,t_{\text{as}}} \cdot c, m \in M \tag{35}$$

where $\eta_1 = 1 + (m-1) \cdot n_s$ and $\eta_2 = \min(m \cdot n_s, T)$.

It is worth mentioning that the proposed algorithm combines the idea of [28] to generate subaperture images, thereby improving the computational efficiency. Each subimage has a full range resolution and low azimuth resolution.

B. Calculation of Complex Fitting Coefficients

To approximate the MOCO function for each frame, it is necessary to calculate the complex fitting coefficients required for each subaperture image in every frame.

The overall residual motion error compensation function corresponding to multiple frames can be expressed as

$$\tilde{f}_{\text{error}}(k) = \exp\left(j4\pi \cdot \frac{\zeta_q \cdot (t_a(k))^2}{\lambda}\right), k \in [1, M \cdot n_s]. \tag{36}$$

Fig. 5 illustrates the construction process of the interframe motion error compensation function. The compensation function for each frame depends not only on the number of pulses

$$F = 1 - \frac{\sum_{i=1}^{x_n}\sum_{j=1}^{y_n}\left(\hat{I}_1(i, j) - m1\right) \cdot \left(\hat{I}_2(i, j) - m2\right)}{\sqrt{\sum_{i=1}^{x_n}\sum_{j=1}^{y_n}\left(\hat{I}_1(i, j) - m1\right)^2} \cdot \sqrt{\sum_{i=1}^{x_n}\sum_{j=1}^{y_n}\left(\hat{I}_2(i, j) - m2\right)^2}} \tag{22}$$

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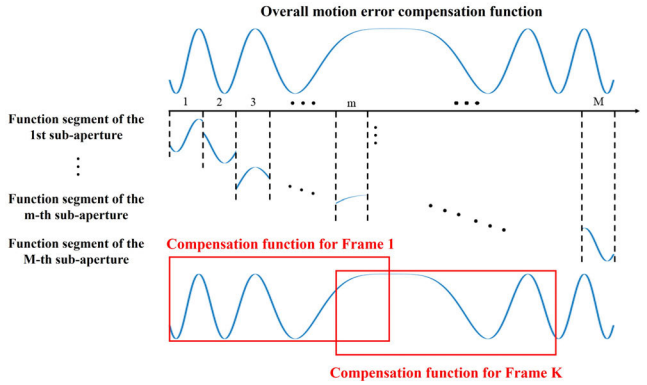


Fig. 5. Construction diagram of interframe motion error compensation function.

contained within the frame but also on the first and last positions of the interframe pulses. To accurately compensate for the motion-induced phase errors, the corresponding segment is extracted from the overall compensation function based on the pulse position. This process ensures that the compensation function of each frame reflects its actual motion state, thereby effectively correcting the interframe image drift caused by phase errors.

Considering that sidelobe interference may obscure weak scattering targets [30], it is essential to apply windowing to subapertures in multiframe video SAR imaging to eliminate the impact of sidelobes on image quality [30], [32].

Unlike the motion error compensation function, the construction of the window function is only related to the number of pulses in each frame. If the pulse count is consistent across frames, the window function can be constructed once and reused on each successive video SAR frame. This feature not only simplifies the construction of the window function but also ensures consistent sidelobe suppression for different frames of images [41], [42]. Using the Hanning window as an example, the required window function $f_{\text{win}}(k')$ is constructed as follows:

$$f_{\text{win}}(k') = 0.5 - 0.5\cos\left(\frac{2\pi k'}{N \cdot n_s}\right), k' \in [1, N \cdot n_s]. \quad (37)$$

Based on (36) and (37), the MOCO window function for the Kth frame can be written as

$$\begin{aligned}\tilde{f}_K(\beta) &= \tilde{f}_{\text{error}}(\alpha) \times f_{\text{win}}(k') \\ &\alpha \in [1+(K-1)\cdot N_2 \cdot n_s, \min(N \cdot n_s + (K-1)\cdot N_2 \cdot n_s, T)] \\ &\beta \in [1, N \cdot n_s]; k' \in [1, N \cdot n_s]; K \in [1, N_f]. \end{aligned} \quad (38)$$

The subfunction segment corresponding to the $\hat{m}$th subaperture of the Kth frame is

$$\begin{aligned}\tilde{f}_{Ks}(\gamma) &= \tilde{f}_K(\hat{\beta}), \hat{\beta} \in [1+(\hat{m}-1)\cdot n_s, \min(\hat{m}\cdot n_s, N \cdot n_s)] \\ &\gamma \in [1, n_s], \hat{m} \in [1, N]. \end{aligned} \quad (39)$$

It should be emphasized that (39) can be multiplied with the subaperture echo to achieve phase error compensation and sidelobe suppression. However, due to changes in pulse positions within each frame, it must slide with the data used to form a frame of image. Therefore, each video frame must be processed separately, leading to repeated calculation in overlapping region of video SAR [35].

From the analysis in Section III-A, each sub-function segment can be approximated using quadratic polynomial fitting. The fit result of (39) can be expressed as follows:

$$\tilde{\text{fit}}_{Ks} = C(\hat{m}, K) \cdot c + B(\hat{m}, K) \cdot b + A(\hat{m}, K) \cdot a \quad (40)$$

where $A(\hat{m}, K)$, $B(\hat{m}, K)$, and $C(\hat{m}, K)$ represent the constant, linear, and quadratic fitting coefficients required for the $\hat{m}$th subaperture in the Kth frame, respectively. It is worth mentioning that the MOCO window functions to be fit are all complex exponential functions, so the fitting coefficients are complex numbers.

The fitting coefficients are crucial in determining the shape, curvature, and variation trend of the fit function. The proposed method calculates these coefficients for each subfunction segment within each frame and adjusts them to control the contribution of weighted subimages, enabling precise fitting of MOCO window segments across frames in the same subaperture. This method fundamentally only modifies the fitting coefficients used for piecewise polynomial approximation, so it does not increase the additional computational burden, thus avoiding redundant calculation of overlapping between frames.

Note that the values of the MOCO window function at the middle and boundaries of each subaperture in different frames are used as complex fitting factors. Since each frame contains N subapertures, a total of $2N + 1$ fitting factors are generated, which serve as constraints for the aperture weighting approximation.

According to (39), the fitting factors required for the $\hat{m}$th subaperture of the Kth frame are

$$\begin{aligned}W_{2\hat{m}-1} &= \tilde{f}_{Ks}(1); W_{2\hat{m}} = \tilde{f}_{Ks}(n_s/2); W_{2\hat{m}+1} \\ &= \tilde{f}_{Ks}(n_s); \hat{m} \in (1, N). \end{aligned} \quad (41)$$

Then, the complex fitting coefficients $A(\hat{m}; K)$, $B(\hat{m}; K)$, and $C(\hat{m}; K)$ can be calculated based on (41) as follows:

$$A(\hat{m}; K) = W_{2\hat{m}-1} \quad (42)$$

$$B(\hat{m}; K) = 4 \cdot W_{2\hat{m}} - 3 \cdot W_{2\hat{m}-1} - W_{2\hat{m}+1} \quad (43)$$

$$C(\hat{m}; K) = 2 \cdot W_{2\hat{m}-1} + 2 \cdot W_{2\hat{m}+1} - 4 \cdot W_{2\hat{m}}. \quad (44)$$

In this article, the detailed derivation process of (42)–(44) is provided in Appendix D.

C. Interframe Combination of Subaperture Images

The next key step in the video SAR process is the subaperture combination. In this step, a set of subaperture images is assigned complex fitting coefficients and subjected to azimuth interpolation. The full-resolution image of each frame is then obtained by summing the subimages. For the echo data corresponding to multiple frames, the index $\xi(K)$ of the first subaperture in the Kth frame image is defined as

$$\xi(K) = (K-1) \cdot N_2. \quad (45)$$

Fig. 6 illustrates the interframe subaperture weighted combination diagram. The per-pulse weights (a, b, c) are applied to each subaperture in different frames to generate three

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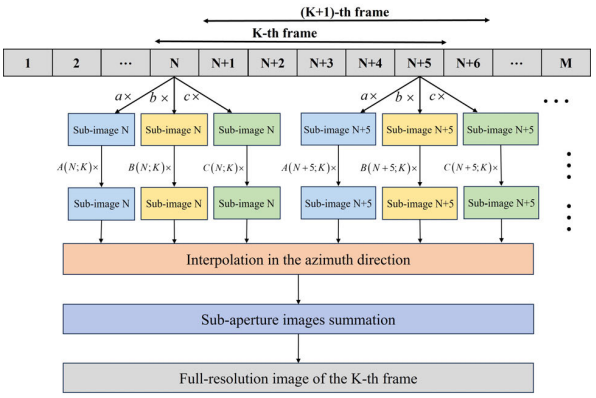


Fig. 6. Interframe subaperture weighted combination diagram.

separate images, respectively. For the Kth frame, the motion error compensation window function is constructed based on the first and last positions of the pulse, and the complex fitting coefficients (A, B, C) for each subaperture within the frame are computed. The fitting coefficients are assigned to the subaperture images corresponding to the constant, linear, and quadratic per-pulse weights. By summing the weighted subaperture images, the final image $\mathrm{IM}_K(i, j)$ of the Kth frame after phase compensation and windowing is obtained as

$$\begin{aligned}
&\mathrm{IM}_K(i, j)\\
&= \sum_{\hat{m}=1}^{N} A\left(\hat{m}; K\right)\cdot \mathrm{IM}_a\left(i, j, m'\right) + \sum_{\hat{m}=1}^{N} B\left(\hat{m}; K\right)\cdot \mathrm{IM}_b\left(i, j, m'\right)\\
&\quad + \sum_{\hat{m}=1}^{N} C\left(\hat{m}; K\right)\cdot \mathrm{IM}_c\left(i, j, m'\right)\\
&= \sum_{\hat{m}=1}^{N} A\left(\hat{m}; K\right)\cdot \sum_{t_{\mathrm{as}}=t_a(\mu_1)}^{t_a(\mu_2)} s_{\mathrm{os}}\left(\mathrm{ID}_{ij,t_{\mathrm{as}}}, t_{\mathrm{as}}\right)\cdot \phi_a(t_{\mathrm{as}})\\
&\quad + \sum_{\hat{m}=1}^{N} B\left(\hat{m}; K\right)\cdot \sum_{t_{\mathrm{as}}=t_a(\mu_1)}^{t_a(\mu_2)} s_{\mathrm{os}}\left(\mathrm{ID}_{ij,t_{\mathrm{as}}}, t_{\mathrm{as}}\right)\cdot \phi_b(t_{\mathrm{as}})\\
&\quad + \sum_{\hat{m}=1}^{N} C\left(\hat{m}; K\right)\cdot \sum_{t_{\mathrm{as}}=t_a(\mu_1)}^{t_a(\mu_2)} s_{\mathrm{os}}\left(\mathrm{ID}_{ij,t_{\mathrm{as}}}, t_{\mathrm{as}}\right)\cdot \phi_c(t_{\mathrm{as}}) \qquad (46)
\end{aligned}$$

where $m' = \xi(K) + \hat{m}$, $\mu_1 = \xi(K)\cdot n_s + (\hat{m}-1)\cdot n_s + 1$, and $\mu_2 = \min((\xi(K) + \hat{m})\cdot n_s, T)$.

From (46), the fit MOCO window function $\mathrm{fit}(\chi)$ for frame K can be expressed as follows:

$$\begin{aligned}
&\mathrm{fit}(\chi)\\
&= \sum_{\hat{m}=1}^{N} A\left(\hat{m}; K\right)\cdot a + \sum_{\hat{m}=1}^{N} B\left(\hat{m}; K\right)\cdot b + \sum_{\hat{m}=1}^{N} C\left(\hat{m}; K\right)\cdot c\\
&= \sum_{\hat{m}=1}^{N} A\left(\hat{m}; K\right)\cdot \left(\frac{n_{\mathrm{re}}-1}{n_s-1}\right)^0 + \sum_{\hat{m}=1}^{N} B\left(\hat{m}; K\right)\cdot \left(\frac{n_{\mathrm{re}}-1}{n_s-1}\right)^1\\
&\quad + \sum_{\hat{m}=1}^{N} C\left(\hat{m}; K\right)\cdot \left(\frac{n_{\mathrm{re}}-1}{n_s-1}\right)^2 \qquad (47)
\end{aligned}$$

where $\chi \in [1, N\cdot n_s]$; $n_{\mathrm{re}} \in [1, n_s]$.

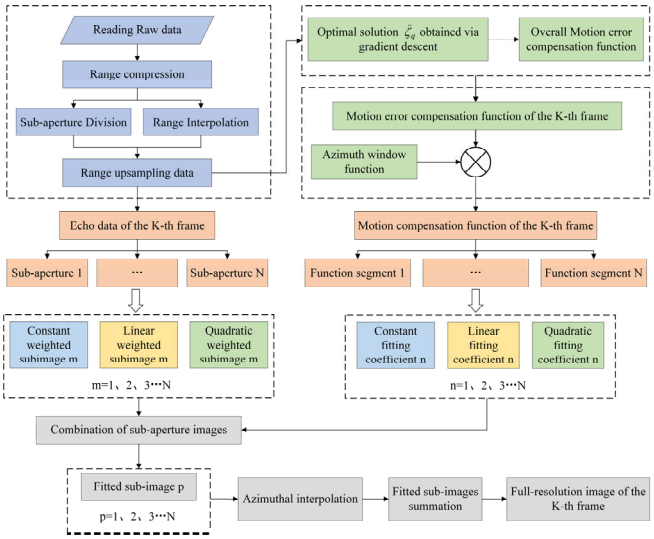


Fig. 7. Framework for integrated fast imaging and stabilization algorithm.

This method achieves multiframe stabilization and sidelobe suppression while avoiding repeated calculations of overlapping pulses between frames, thereby enhancing the efficiency of video SAR imaging.

IV. Integrated Imaging and Stabilization Processing Method

A. Integrated Process

The overall framework of the proposed integrated method for video SAR stabilization and fast imaging is shown in Fig. 7. The framework consists of interframe motion error estimation module and SCFBP module.

The interframe motion error estimation module divides the echo data into subapertures and introduces a gradient function. The quadratic coefficient of the slant range error is accurately estimated using the gradient descent method by maximizing the correlation of subaperture images.

The SCFBP module generates three separate subimages for any subaperture based on the different weights of the pulse contribution. Then, the MOCO window function for the current frame is constructed using the estimated quadratic coefficient, and the complex fitting coefficients corresponding to each subaperture are calculated. By assigning these fitting coefficients to subaperture images with constant, linear, and quadratic pulse weights, the weighted combination of the subimages achieves interframe high stability and fast imaging in video SAR. The detailed steps of the integrated algorithm are summarized in Table I.

B. Operational Analysis

In this study, the SCFBP module is incorporated into the integrated framework to avoid redundant computations caused by interframe overlap during multiframe MOCO and windowing, thereby improving imaging efficiency. Next, the computational costs of traditional BP, FBP, and SCFBP in multiframe imaging are compared and analyzed.

As mentioned above, the total number of frames is N_f, and each frame contains N subapertures. N_1 and N_2 are

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TABLE I
INTEGRATED FAST IMAGING AND STABILIZATION ALGORITHM

Algorithm 1 Integrated Fast Imaging and Stabilization Algorithm.

Input: Raw Data, Inertial Navigation System Data

Output: A total of K frames of video SAR imaging results.

Preprocess Raw Data:

1. Define inter-frame imaging parameters;
2. Sub-aperture division;
3. Get per-pulse weights a, b, c based on the sub-aperture division;

Motion error estimation process based on object function:

1. Establish objective function F of quadratic coefficient ζ_q based on correlation between sub-images;
2. Define the upper and lower bounds ub and lb;
3. Obtain the objective function with penalty term F';
4. Parameter search operations based on gradient descent method;
5. Get optimal solution $\hat{\zeta}_q$ corresponding to the minimum value of F'.

SCFBP Imaging:

1. Generate the three independent sub-images IM_a, IM_b, IM_c of each sub-aperture in the raw data based on per-pulse weights a, b, c;
2. Get overall motion error compensation function based on $\hat{\zeta}_q$;
3. The total number of frames to be imaged is K, for k in K do;
4. Get the motion compensation window function of the k-th frame;
5. Get the fitting coefficients A, B, C of the k-th frame based on least square method;
6. Get the sub-images of the k-th frame and give A, B, C to the sub-images with different per-pulse weights;
7. Get the k-th frame image by combining and summing these sub-images;

the number of overlapping and nonoverlapping subapertures between frames. Each subaperture contains n_s pulses. The size of the imaging grid is $x_n \times y_n$. Both FBP and SCFBP first compute subaperture images on a coarse azimuth pixel grid and then apply azimuth interpolation to obtain full-resolution images. Let the azimuth interpolation factor for subapertures be σ, and $\sigma > 1$.

For traditional BP, after the MOCO and windowing, the number of operations for each subaperture imaging D_1 is

$$D_1 = x_n \cdot y_n \cdot n_s. \tag{48}$$

The number of operations D_2 to generate a subaperture image in the FBP and SCFBP can be written as

$$D_2 = \frac{x_n}{\sigma} \cdot y_n \cdot n_s. \tag{49}$$

When using the traditional BP for multiframe imaging, each subaperture in every frame requires MOCO and windowing, so the total operations count D_t of the traditional BP is

$$D_t = N_f \cdot N \cdot D_1 = N_f \cdot N \cdot x_n \cdot y_n \cdot n_s. \tag{50}$$

In the FBP, low-resolution imaging is performed on each subaperture, and the computational efficiency is improved by coherently summing the subaperture images. Consequently, the total operations count D_f of the FBP is

$$D_f = N_f \cdot N \cdot D_2 = N_f \cdot N \cdot \frac{x_n}{\sigma} \cdot y_n \cdot n_s. \tag{51}$$

TABLE II
MAIN PARAMETERS OF SIMULATED DATA

Parameter	Value
Wavelength	Millimeter wave
Signal bandwidth	300MHz
Platform velocity	70m/s
Platform altitude	3km
Point T1 coordinate	(0,0,0)
Point T2 coordinate	(7,0,0)
Point T3 coordinate	(-7,0,0)
Point T4 coordinate	(0,7,0)
Point T5 coordinate	(0,-7,0)

The proposed SCFBP generates three separate subimages for each subaperture contained in the entire echo data. By assigning different complex fitting coefficients to these subimages and performing azimuth interpolation, the subimages required for each frame are combined and summed, thereby avoiding the redundant calculation of interframe overlapping regions. Therefore, the total operations count D_w of SCFBP is

$$\begin{aligned} D_w &= 3 \cdot \left(N + \left(N_f - 1\right) \cdot N_2\right) \cdot D_2 \\ &= 3 \cdot \left(N + \left(N_f - 1\right) \cdot N_2\right) \cdot \frac{x_n}{\sigma} \cdot y_n \cdot n_s. \end{aligned} \tag{52}$$

According to (50) and (52), the difference in the number of operations between traditional BP and SCFBP for multiframe imaging is denoted as ΔD_{tw} and is given by

$$\begin{aligned} \Delta D_{\text{tw}} &= D_t - D_w \\ &= \left(\left(\sigma \cdot N_f - 3\right) \cdot N - 3 \cdot \left(N_f - 1\right) \cdot N_2\right) \cdot D_2. \end{aligned} \tag{53}$$

By comparing (51) and (52), the difference in the number of operations between the FBP and SCFBP is ΔD_{fw}

$$\begin{aligned} \Delta D_{\text{fw}} &= D_f - D_w \\ &= \left(\left(N_f - 3\right) \cdot N - 3 \cdot \left(N_f - 1\right) \cdot N_2\right) \cdot D_2. \end{aligned} \tag{54}$$

The high-frame-rate characteristic of video SAR means $N_f \gg 3$. Therefore, (53) and (54) can be approximated as

$$\Delta D_{\text{tw}} = (\sigma \cdot N - 3 \cdot N_2) \cdot N_f \cdot D_2 \tag{55}$$

$$\Delta D_{\text{fw}} = (N - 3 \cdot N_2) \cdot N_f \cdot D_2. \tag{56}$$

Equations (55) and (56) indicate that when $N > 3 \cdot N_2$, SCFBP yields a lower computational cost compared to traditional BP and FBP. This condition is typically satisfied in video SAR applications with a high interframe overlap rate.

V. SIMULATION

To validate the effectiveness of the proposed method, some numerical experiments are first conducted using simulated millimeter-wave video SAR data. Some key system parameters and point target coordinates are listed in Table II. Motion errors are easily introduced during the collection of echo signal due to the influence of atmospheric turbulence. These slight errors are imperceptible to the inertial navigation system (INS) and result in residual quadratic phase errors between frames, which subsequently cause point target defocusing and drift across multiframe images. Therefore, it should be

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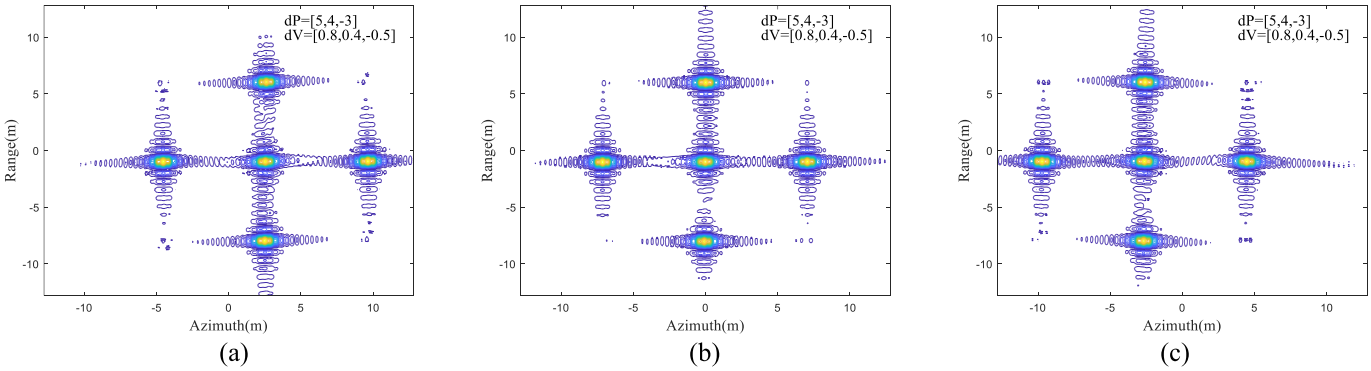


Fig. 8. Imaging results of the point targets without interframe MOCO and windowing. (a)–(c) Results in 1st, 21st, and 41st frames, respectively.

TABLE III
INTERFRAME IMAGING PARAMETERS

Parameter	Value
Image size	512×512
Total sub-apertures	512
Sub-apertures per frame	112
Inter-frame overlapping sub-apertures	102
Grid resolution	0.05m
Imaging frames	41
Pulses per frame	1792
Inter-frame overlap rate	91%

clarified that, in Sections V and VI, inter-frame MOCO specifically refers to the compensation of such residual interframe quadratic phase errors in video SAR.

In this simulation, the initial position error of the platform is set to [4, 5, −3] m, and the initial velocity error is set to [0.4, 0.8, −0.5] m/s. These errors correspond to the range, azimuth, and altitude dimensions and are assumed to be fixed values. All experiments are carried out on a personal computer running Windows 10, with Intel Core i7-975H at 2.6 GHz and 16-GB RAM.

A. Comparison and Analysis of Imaging Performance

Interframe MOCO and windowing are applied to the simulated data using traditional BP, FBP, and the proposed SCFBP. Furthermore, the performance of these methods in terms of imaging efficiency and stabilization accuracy is compared and analyzed in detail. Table III shows the interframe simulation parameters. The data contain 8192 pulses, which are divided into 512 subapertures. The interframe overlap rate is 91%, corresponding to 41 frames of radar images with an imaging resolution of 0.05 m.

Due to space constraints, the results of Frames 1, 21, and 41 are selected for analysis. Fig. 8 presents the simulated point targets without interframe MOCO and windowing. Comparing these frames reveals that motion errors cause obvious interframe drift and defocusing along the azimuth direction. The cumulative drift error is 52 pixels between the 1st and 21st frames and 104 pixels between the 1st and 41st frames. Furthermore, each point target has a high sidelobe in both the range and azimuth directions.

To achieve interframe image stabilization in video SAR, it is necessary to estimate the motion error between frames. In this simulation, the optimal quadratic coefficient of the slant range error, estimated using the gradient descent method, is 0.011. Based on this coefficient, the MOCO function for multiple frames is subsequently constructed.

Windowing is utilized for isolating the desired band of frequency and zero-valuing the signal outside that interval [41]. To enhance the image quality, sidelobe suppression can be achieved by introducing the window function into the BP algorithm during the imaging process [43]. Fig. 9 presents the imaging results using traditional BP and FBP in Frames 1, 21, and 41, where MOCO and windowing are applied in each frame.

As shown in Fig. 9, the point targets exhibit a well-focused state. Moreover, the targets in different frames are corrected to the same position within the scene, effectively addressing the issue of interframe image drift. In addition, the application of windowing achieves consistent sidelobe suppression for all point targets across multiple frames.

The proposed SCFBP method fully exploits the interframe overlap of video SAR. By dividing the multiframe echo data into subapertures with equal pulse interval [44], it ensures that subapertures in different frames can use a consistent set of per-pulse weights (a, b, c) to generate subimages. Subsequently, MOCO window functions specific to each frame are constructed, and the complex fitting coefficients (A, B, C) required for each subaperture within each frame are calculated. These fitting coefficients are then assigned to the corresponding subimages, and the final image of each frame is obtained by performing a weighted summation of the subaperture images.

The multiframe imaging results obtained by the SCFBP are shown in Fig. 10. It is evident that the issue of interframe image drift has been solved, and the sidelobes of point targets are suppressed.

For detailed evaluation, the azimuth profiles of point target T1 in the 21st frame from different imaging approaches are plotted in Fig. 11 and the corresponding performance values are listed in Table IV, including peak sidelobe ratio (PSLR), integrated sidelobe ratio (ISLR), and impulse response width (IRW). Additionally, the cumulative mean drift error and the total imaging time for all 41 frames are also provided in Table IV. The results indicate that, without interframe MOCO

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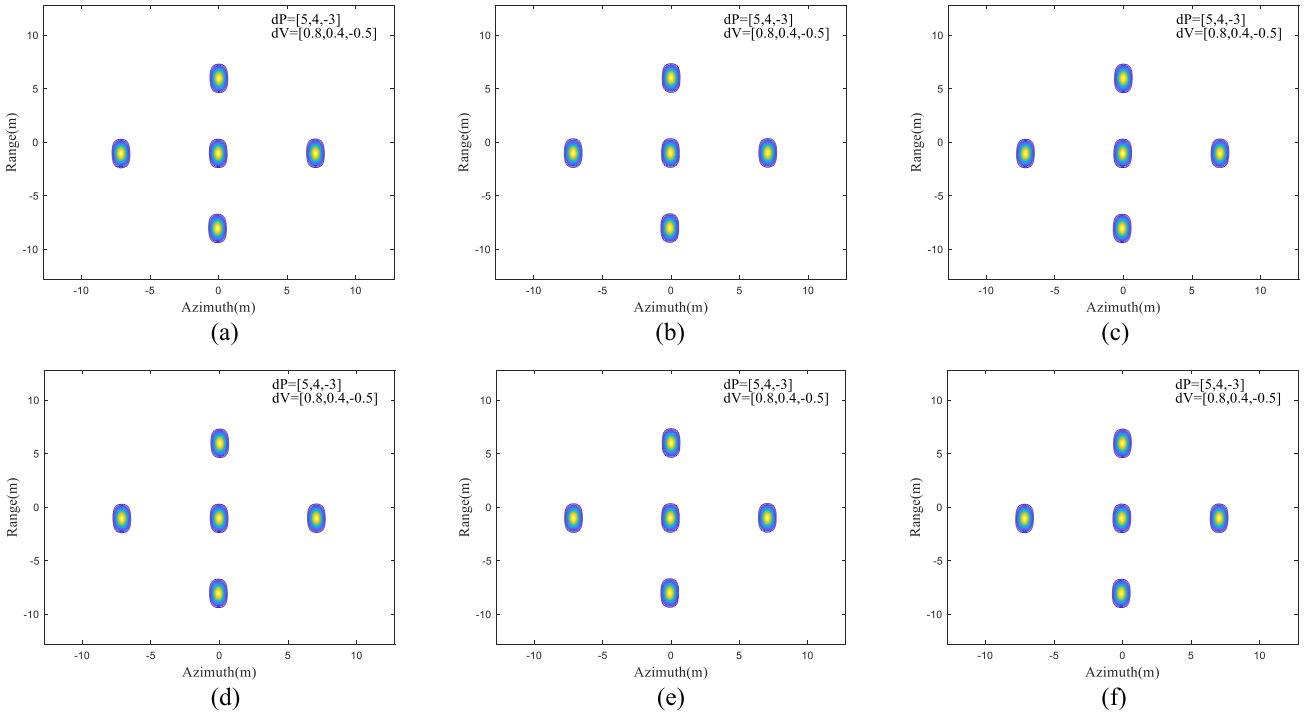


Fig. 9. Imaging results of the point targets processed by traditional BP and FBP with interframe MOCO and windowing. (a)–(c) Results processed by traditional BP in 1st, 21st, and 41st frames, respectively. (d)–(f) Results processed by FBP in 1st, 21st, and 41st frames, respectively.

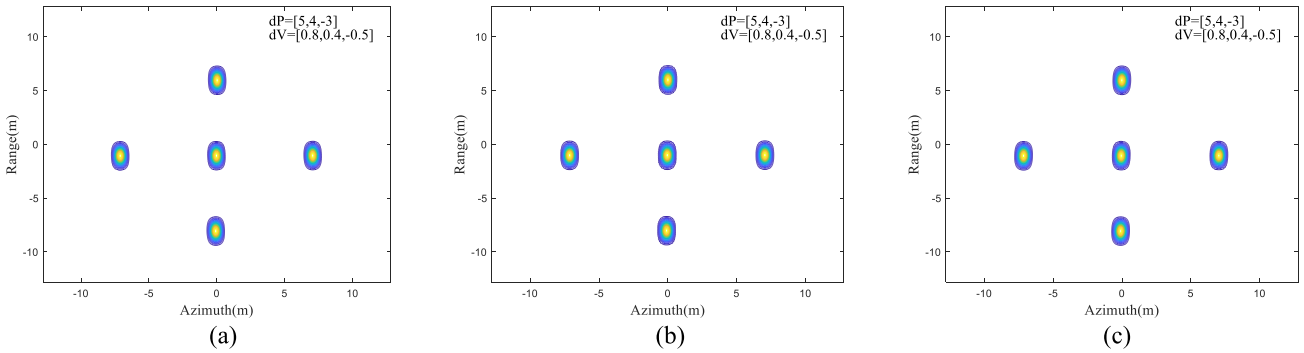


Fig. 10. Imaging results of the point targets processed by SCFBP. (a)–(c) Results in 1st, 21st, and 41st frames, respectively.

TABLE IV
PERFORMANCE VALUE COMPARISON OF DIFFERENT APPROACHES

Method	IRW (m)	PSLR (dB)	ISLR (dB)	Cumulative drift error (pixel)	Runtime (s)
Traditional BP	0.49	-1.49	1.32	104	10120
Traditional BP+MOCO+Win	0.48	-43.5	-43.65	0	10707
FBP+MOCO+Win	0.48	-44.9	-44.4	2	5363
SCFBP	0.48	-45.1	-41.95	0	1596

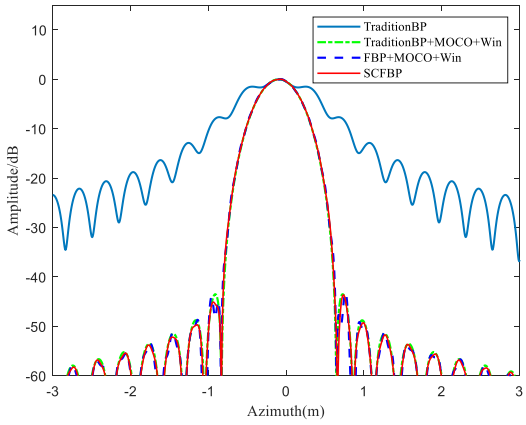


Fig. 11. Azimuth profiles from different imaging approaches of T1.

and windowing during the imaging process, the target fails to be well-focused, and the drift amount accumulated in 41 frames reaches 104 pixels. Moreover, the traditional BP is extremely time-consuming for multiframe imaging due to its high computational load.

Interframe MOCO in traditional BP can effectively address point target defocusing and multiframe image drift. As a result, the cumulative drift over the 41 frames is reduced to zero. The Hanning window is selected for sidelobe suppression. According to the quality parameters in Table IV, the PSLR and ISLR values are greatly reduced after windowing, indicating

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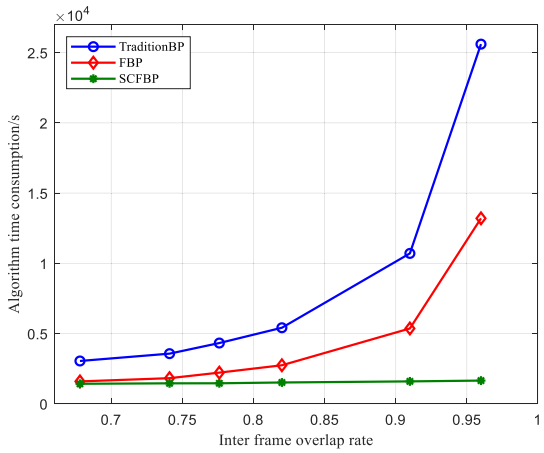


Fig. 12. Comparison of runtime among traditional BP, FBP, and SCFBP.



Fig. 13. Imaging result of millimeter-wave airborne video SAR.

that sidelobe interference is eliminated. However, redundant computations in the overlapping regions between consecutive frames still lead to increased imaging time.

Interframe MOCO and windowing are also applicable to FBP. Compared with traditional BP, FBP improves computational efficiency by performing low-resolution imaging on each subaperture. Nevertheless, it still suffers from redundant computations during multiframe imaging.

The total runtime of SCFBP for processing 41 frames is only 1596 s. The proposed method achieves comparable image quality and stabilization accuracy to those of traditional BP and FBP. Furthermore, the multiframe imaging speed is significantly improved.

B. Analysis of the Acceleration Performance of SCFBP

Next, under the condition of a fixed total number of subapertures, the effect of different aperture overlap rates on the acceleration performance of SCFBP is analyzed. In this simulation, the echo data are divided into 512 subapertures, with each frame containing 112 subapertures. Table V presents the runtimes of traditional BP, FBP, and SCFBP under various aperture overlap rates and provides the corresponding acceleration efficiencies of SCFBP under equal conditions.

Fig. 12 illustrates the runtime comparison under different interframe overlap rates. Given a fixed total number of subapertures, a higher interframe overlap rate results in more imaging frames. During interframe MOCO and windowing in traditional BP or FBP, the number of repeated calculations for overlapping between frames increases accordingly, thereby imposing a greater computation burden.

The increase of interframe overlap rate raises the usage count of per-pulse weights (a, b, c) in SCFBP, thereby reducing the time required for multiframe imaging. This result aligns with the computational analysis presented earlier in this article. To achieve imaging at higher frame rates, video SAR typically requires a high interframe overlap rate. For example, the aperture overlap rate reaches up to 90% in [1] and [45]. Therefore, the imaging efficiency of SCFBP is significantly superior to that of traditional BP and FBP under conditions of high interframe overlap.

VI. Experiments on Real Data

In the following, the real data are used to investigate the effectiveness of the proposed method. The system and image parameters of ViSAR are presented in Table VI. The dataset is collected by a millimeter-wave video SAR system mounted on the testing aircraft. The video SAR works in spotlight mode and flies along a circular trajectory. Owing to the accuracy limitation of INS and the influence of atmospheric turbulence, slight motion errors exist between the real and measured trajectories.

The detailed analyses are organized as follows. First, the proposed SCFBP is applied to generate multiframe ViSAR images to demonstrate its effectiveness in drift elimination and sidelobe suppression. Then, the image stabilization accuracy of SCFBP is quantitatively compared with that of the widely used SIFT-based registration method, and their differences in robustness and computational complexity are analyzed. Finally, comparisons among different imaging methods are conducted to evaluate imaging quality and acceleration performance, thereby validating the superiority of the proposed method in fast and stable multiframe imaging.

A. ViSAR Image Generation Using SCFBP

Fig. 13 shows the imaging result of the raw data using traditional BP without MOCO and windowing. Although the image is visually focused, slight blurring becomes noticeable upon magnified observation. To illustrate the details of the proposed SCFBP method, three local images are marked with rectangles in the figure, and the 1st, 41st, 81st, and 121st frames are selected for comparative analysis. The size of individual local image is set as 128 × 128 (range × azimuth) pixels. It is worth mentioning that the selected local images contain a large amount of strong points, cover the most dispersed power, and distribute over a wide extent [40].

By accurately estimating the quadratic coefficient of slant range error, a motion error compensation function can be constructed to correct interframe phase error. The motion error caused by atmospheric turbulence is generally small, with the corresponding quadratic coefficient typically not exceeding the order of 0.1. In this study, the prior information of motion error

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TABLE V
RUNNING TIME AND ACCELERATION PERFORMANCE WITH DIFFERENT INTERFRAME OVERLAP RATES

Inter-frame overlap rate	Total sub-apertures	Runtime			Acceleration performance	
		Traditional BP +MOCO+Win	FBP+MOCO+Win	SCFBP	Traditional BP +MOCO+Win	FBP+MOCO+Win
67.8%	512	3049s	1594s	1426s	53.2%	10.5%
74.1%	512	3571s	1825s	1458s	59.2%	20.1%
77.6%	512	4326s	2220s	1461s	66.2%	34.2%
82%	512	5415s	2744s	1518s	71.9%	44.7%
91%	512	10707s	5363s	1596s	84.6%	70.2%
96%	512	25605s	13203s	1653s	93.5%	87.5%

TABLE VI
MAIN PARAMETERS OF REAL DATA

Parameter	Value
Wavelength	Millimeter wave
Signal bandwidth	900MHz
Mean velocity	69m/s
Platform flight radius	665m
Mean altitude	2.5km
Image grid (range×azimuth)	0.3m×0.3m
Image size (range×azimuth)	1024×1024
Aperture overlapped rate	87.5%

is utilized to accelerate convergence by introducing a penalty term, which effectively reduces the time cost of parameter search. The optimal quadratic coefficient value obtained via the gradient descent method is 0.0054. Based on this estimated coefficient, MOCO functions are constructed for each frame and combined with a window to generate the required MOCO window functions.

For the selected frames (1st, 41st, 81st, and 121st), piecewise polynomial fitting methods of different orders are employed to approximate the MOCO window function. The comparisons of the fitting results are shown in Fig. 14(a)–(d). As shown in the figure, the theoretical function, given by (38), represents the MOCO window function of the current frame before piecewise fitting. The fit functions correspond to the approximations of the theoretical function using piecewise constant, linear, and quadratic models.

It can be observed that the similarity between the theoretical and fit functions gradually improves as the polynomial order increases. The quadratic fitting function closely matches the theoretical function. Therefore, although the order of the polynomial can be arbitrary, the quadratic approximation of the aperture weighting is adopted in this article to balance the imaging speed, imaging stability, and sidelobe performance, thereby achieving the best comprehensive effect.

Taking local image 1 as an example, the first row of Fig. 15 presents the imaging results of the 1st, 41st, 81st, and 121st frames before SCFBP processing. By comparing the images across frames, interframe drift can be clearly observed, with the drift accumulating as the number of frames increases. Moreover, the absence of windowing during the imaging process results in the leakage of main lobe energy to the sides.

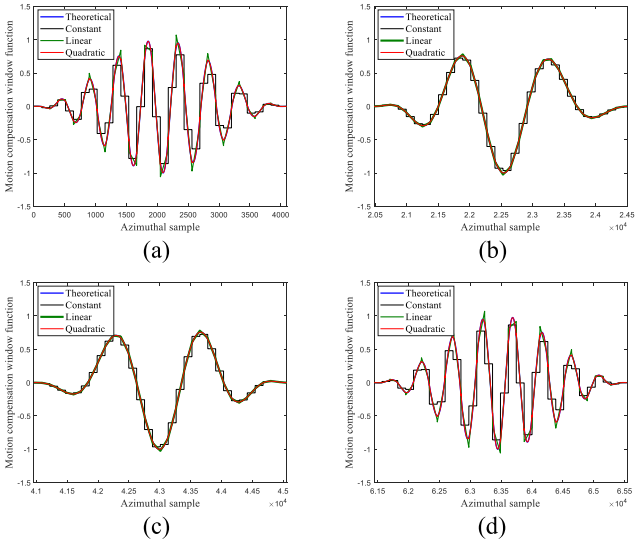


Fig. 14. Comparison of piecewise constant, linear, and quadratic fitting functions for different frames. (a) Frame 1. (b) Frame 41. (c) Frame 81. (d) Frame 121.

Fig. 15(a)–(c) presents the multiframe imaging results of different local images before and after using SCFBP. By compensating the phase error in each frame, the issue of interframe image drift can be effectively solved, which also validates the accuracy of the parameter estimation. In addition, the sidelobes of the target are suppressed, which makes the image more concentrated and achieves better focusing performance.

B. Comparison and Analysis of Stabilization Performance

As shown in Fig. 16, two control points in the observation scene are selected to evaluate the enhancement in image stability. The stabilization accuracy is quantitatively assessed by measuring the pixel offsets of these points in the X- and Y-directions before and after algorithm processing. For comparison and analysis, both SIFT and the proposed SCFBP method are used to register the collected multiframe raw video SAR images.

Given a sensed image and a reference image, the objective of registration is to discover a correspondence (transformation) between both images such that the transferred image is as similar as possible to the reference image [16], [46]. The core idea of the SIFT is to detect the extreme points of the

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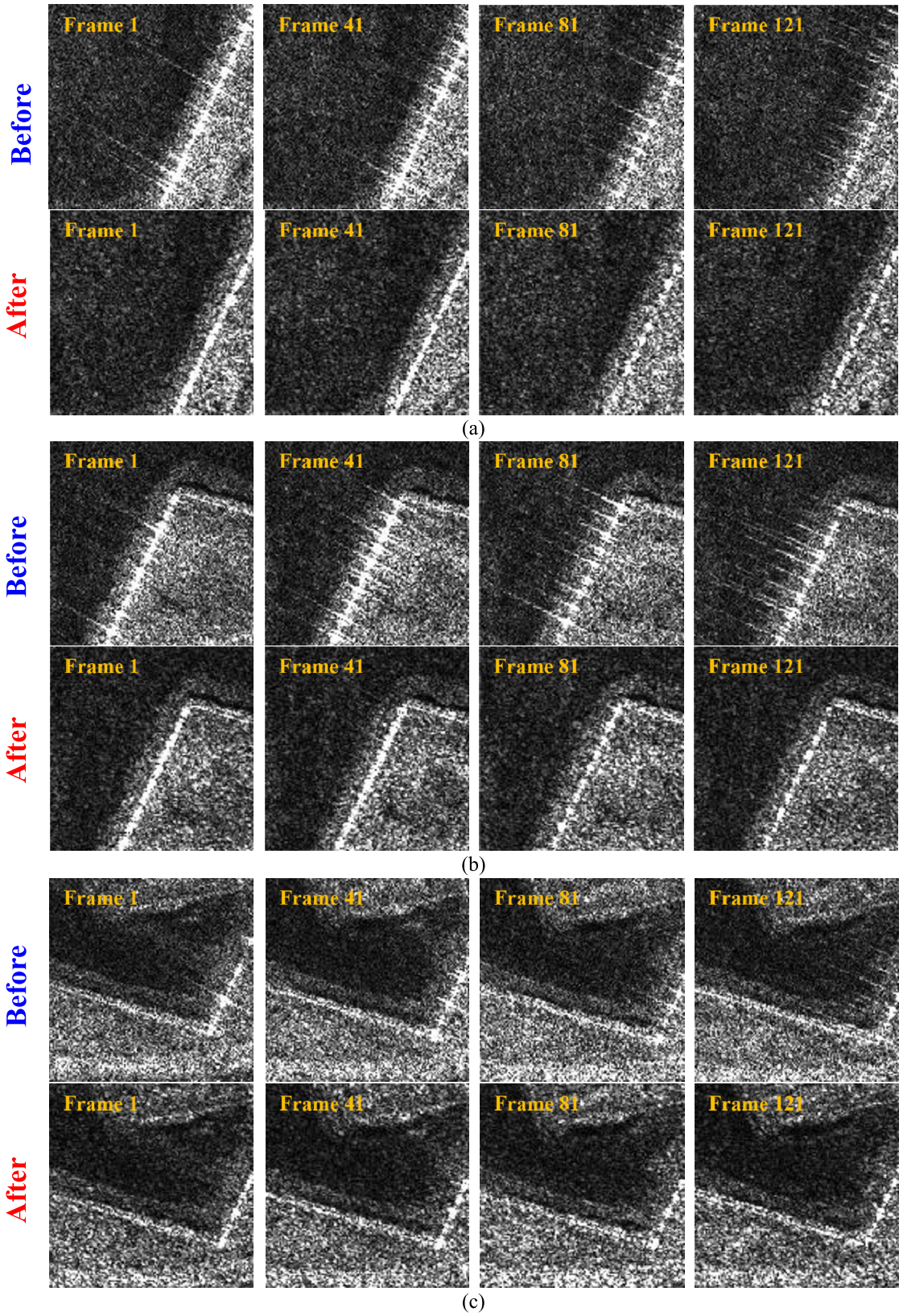


Fig. 15. Imaging results of the 1st, 41st, 81st, and 121st frames for different local images (top) before and (bottom) after using SCFBP. (a) Local image 1. (b) Local image 2. (c) Local image 3.

image in the scale space and extract the position, scale, and rotation invariant features. The extracted features are described by high-dimensional vectors and matched by calculating the Euclidean distance between these vectors [17], [18], [19].

In this article, multiframe imaging is performed using traditional BP, and SIFT is employed to achieve interframe image registration. A total of 121 frames are generated through traditional BP. The first frame is used as the reference image,

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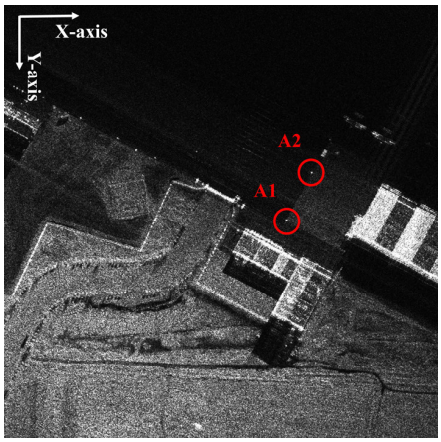


Fig. 16. Distribution of control points A1 and A2.

TABLE VII
EVALUATION OF IMAGE STABILIZATION PERFORMANCE

Control Point	Approach	Error in X direction (pixel)	Error in Y direction (pixel)	Plane error (pixel)
A1	Traditional BP	14	6	15
	Traditional BP +SIFT	2	0	2
	SCFBP	0	1	1
A2	Traditional BP	20	4	20
	Traditional BP +SIFT	1	2	2
	SCFBP	0	0	0

and the subsequent frames are treated as sensed images [15]. SIFT is then applied to register the sensed images with the reference image sequentially. It is worth mentioning that the random sample consensus (RANSAC) method is introduced to remove the outliers during image registration [47], which ensures the robustness of SIFT in feature point matching and increases the probability of correct matching.

Fig. 17 shows the local images of the two control points in the 1st and 121st frames, processed by traditional BP, traditional BP with SIFT registration, and SCFBP. Each local image with a control point contains 128 × 128 (range × azimuth) pixels.

Table VII lists the pixel drift in the X- and Y-directions for different control points between Frames 1 and 121. The results indicate obvious pixel drift between control points without interframe image registration. Both SIFT and SCFBP can effectively reduce the drift, with SCFBP achieving slightly higher stabilization accuracy than SIFT.

SIFT performs registration in the image domain based on the acquired radar images. However, SAR images are affected by a strong multiplicative noise known as speckle, which limits the matching performance of SIFT [14]. The influence of speckle noise on SAR image registration is mainly reflected in two stages: 1) in the feature detection stage, it causes SIFT to detect numerous false feature points, which are highly random and prone to mismatch, affecting the performance of extracting corresponding points [15], [22] and 2) in the feature description stage, descriptors are generated from gradient histograms in the neighborhood of the feature points. Speckle noise alters the statistical specificities of the local gradients, degrading the quality of the descriptor, thereby reducing the distinguishability of the feature points.

Additionally, during the construction of the Gaussian scale space, SIFT employs the difference of Gaussian (DoG) operator, whose edge response can severely degrade the accuracy of SAR image registration. If this response is not eliminated, the detected feature points tend to be concentrated at the edges of the SAR image, where their close neighborhood increases the possibility of mismatches [48].

Compared to SIFT, the proposed SCFBP is not affected by speckle noise or other factors. Its advantage is modeling and estimating interframe drift errors, addressing multiframe image drift through phase compensation in the signal domain.

TABLE VIII
COMPARISON OF RUNNING TIME AND REGISTRATION ACCURACY

Approach	Error in X direction (pixel)	Error in Y direction (pixel)	Plane error (pixel)	Runtime (s)
Traditional BP	17	5	17.7	57259.4s
Traditional BP +SIFT	1.5	1	1.8	59000.4s
SCFBP	0	0.5	0.5	4450.3s

Table VIII presents the multiframe imaging time and the average stabilization accuracy for different methods. The results indicate that, given the same number of frames and comparable stabilization accuracy, traditional BP combined with SIFT requires much more time than SCFBP.

The reasons for this issue are as follows: 1) the redundant computations caused by high interframe overlap increase the overall imaging time in traditional BP and 2) after generating the multiframe images, SIFT is applied to perform image registration on a frame-by-frame basis. SIFT usually consists of four stages: scale-space extrema detection, key point localization, orientation assignment, and key point descriptor generation [18], [46], [49]. Each of these steps requires a lot of calculation.

During the construction of the scale space, SIFT generates images at different scales by performing multiple Gaussian blurs, thereby establishing a Gaussian pyramid. This process involves several convolution operations. Then, the DoG operator is computed between adjacent scale images to detect potential key points. This operation requires multiple computations for each pixel [18], [19].

During feature point matching, each feature point in the reference image is matched to a corresponding point in the sensed image. The Euclidean distance is commonly used to compare feature descriptors, which requires substantial vector operations [20], [21]. In addition, feature descriptors are typically 128-D vectors, and the comparison and matching of high-dimensional feature vectors involve heavy floating-point

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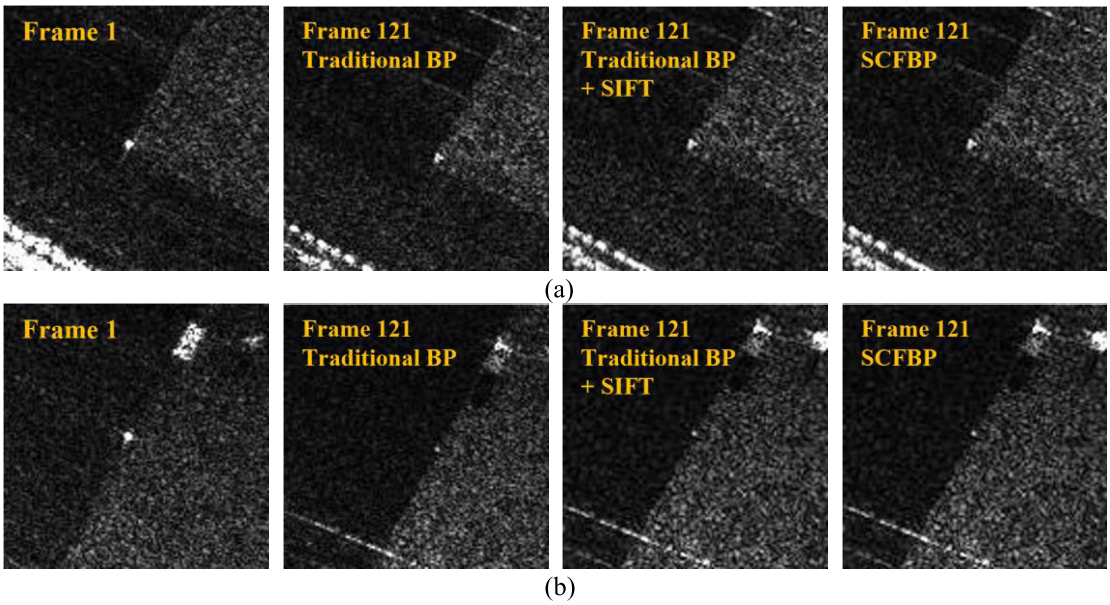


Fig. 17. Comparison of local images of different control points processed by different methods. (a) Control point A1. (b) Control point A2.

TABLE IX
QUALITY PARAMETER COMPARISON FOR FIG. 20

Target	Method	PSLR(dB)	ISLR(dB)	IRW(imaging grids)
A1	Traditional BP	-8.6967	-2.5236	0.3063
	Traditional BP+MOCO	-14.8495	-9.6659	0.3625
	Traditional BP+MOCO+Win	-25.2128	-12.0983	0.4375
	FBP+MOCO+Win	-25.2124	-12.1973	0.4312
	SCFBP	-24.9984	-12.1033	0.4437
A2	Traditioanl BP	-7.9934	-2.4221	0.2688
	Traditional BP+MOCO	-9.9594	-5.8436	0.4313
	Traditional BP+MOCO+Win	-24.8206	-14.0832	0.4250
	FBP+MOCO+Win	-24.9873	-14.0198	0.4063
	SCFBP	-26.9195	-14.2364	0.3937

TABLE X
RUNNING TIME AND ACCELERATION PERFORMANCE WITH DIFFERENT INTERFRAME OVERLAP RATES

Inter-frame overlap rate	Total sub-apertures	Runtime			Acceleration performance	
		Traditional BP +MOCO+Win	FBP+MOCO+Win	SCFBP	Traditional BP +MOCO+Win	FBP+MOCO+Win
75%	4096	25410.7s	15576.4s	4428.7s	82.6%	71.6%
81.25%	4096	36596.2s	21341.3s	4598.2s	87.4%	78.5%
87.5%	4096	61128.5s	35821.1s	4615.0s	92.5%	87.2%
90.625%	4096	80803.2s	51192.0s	4649.6s	94.2%	91.0%
93.75%	4096	12259.3s	74854.6s	4765.4s	96.1%	93.6%
96.875%	4096	247137.8s	153054.2s	5286.6s	97.9%	96.5%

operation, further increasing the computational complexity. Furthermore, to improve matching accuracy, methods such as RANSAC are commonly employed to filter and refine the initial matching results [47]. These methods require iterative computations to remove false matches and identify the optimal registration model, thus adding to the time cost of SIFT in image registration.

C. Comparison and Assessment of Imaging Efficiency

To evaluate the focusing performance, control points A1 and A2 in the scene are processed using different methods, and the results are shown in Figs. 18 and 19. As illustrated, the combination of windowing and MOCO almost completely eliminates motion error disturbance and yields improved focusing quality.

The azimuth profiles of A1 and A2 are plotted in Fig. 20(a) and (b), respectively. The related PSLR, ISLR, and IRW are listed in Table IX. It can be seen that windowing effectively reduces the PSLR and ISLR of point targets but broadens the main lobe, resulting in an increased IRW and degraded azimuth resolution.

Fig. 20 and Table IX demonstrate that the proposed method achieves imaging quality comparable to that of traditional

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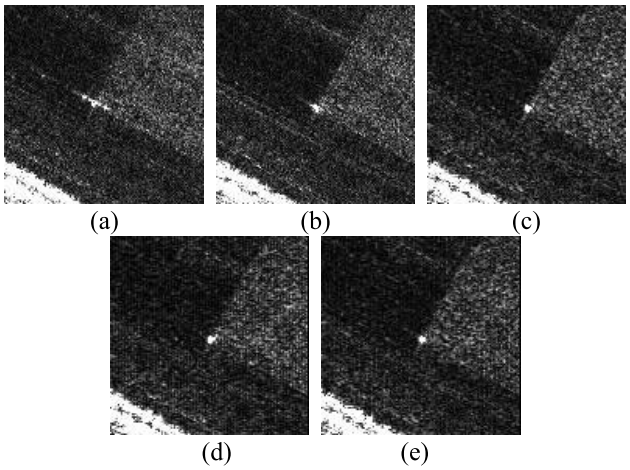


Fig. 18. Local image comparison. (a) Local image of A1 obtained by traditional BP. (b) Traditional BP-based MOCO. (c) Traditional BP-based MOCO and windowing. (d) FBP-based MOCO and windowing. (e) SCFBP.

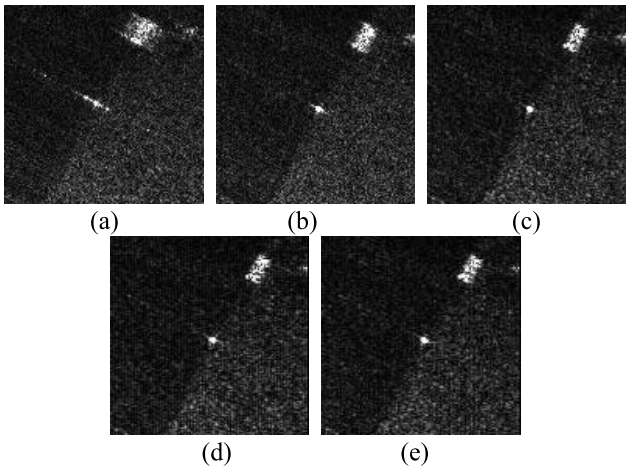


Fig. 19. Local image comparison. (a) Local image of A2 obtained by traditional BP. (b) Traditional BP-based MOCO. (c) Traditional BP-based MOCO and windowing. (d) FBP-based MOCO and windowing. (e) SCFBP.

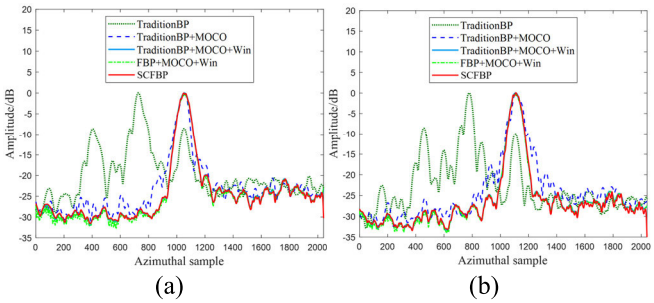


Fig. 20. Azimuth profiles. (a) Control point A1. (b) Control point A2.

BP and FBP with interframe MOCO and windowing. The primary difference among the three methods lies in the time required for multiframe imaging. Therefore, the acceleration performance of SCFBP is analyzed below.

Table X summarizes the imaging time under different aperture overlap rates, and Fig. 21 presents the comparison of time consumption. The results indicate that the proposed method exhibits better imaging acceleration performance, providing a new possibility for fast multiframe imaging in video SAR.

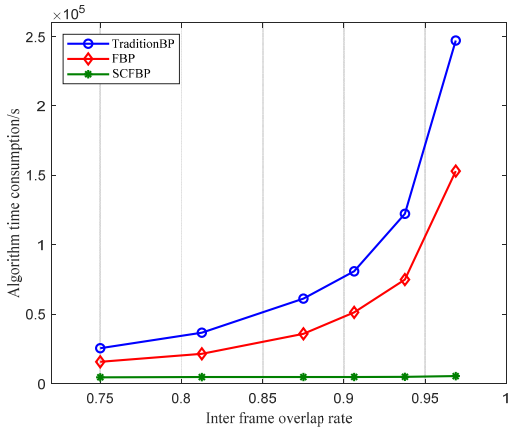


Fig. 21. Comparison of runtime among traditional BP, FBP, and SCFBP.

VII. Conclusion

In this article, an integrated fast multiframe imaging and stabilization method for video SAR based on interframe drift compensation is proposed. The generation mechanism of drift error is analyzed from the imaging aspect. A gradient function is constructed to accurately estimate the motion error by maximizing the correlation between subaperture images. Based on the estimated error parameter, the SCFBP algorithm is presented. Three separate complex-weighted images are generated for each subaperture, which can be combined to approximate any desired motion error compensation function within the frame. Simulations and real data demonstrate that for high frame rate and high overlap millimeter-wave video SAR imaging, the proposed method effectively improves interframe stability and imaging efficiency.

Appendix A

From (3), the expression of $R_e(t_a)$ after performing the Taylor expansion at $t_a = 0$ can be rewritten as follows:

$$R_e(t_a) = R_e(0) + R_e'(0)\cdot t_a + \frac{1}{2}\cdot R_e''(0)\cdot t_a^2 + \frac{1}{6}\cdot R_e^{(3)}(0)\cdot t_a^3 + o\left(t_a^4\right). \tag{A1}$$

The phase error induced by (A1) is

$$\Delta\phi(t_a) = -\frac{4\pi}{\lambda}\cdot\left(R_e(0) + R_e'(0)\cdot t_a + \frac{1}{2}\cdot R_e''(0)\cdot t_a^2 + \frac{1}{6}\cdot R_e^{(3)}(0)\cdot t_a^3 + o\left(t_a^4\right)\right). \tag{A2}$$

The phase error corresponding to the higher order terms is given by

$$\Delta\tilde{\phi}(t_a) = -\frac{4\pi}{\lambda}\cdot\left(\frac{1}{6}\cdot R_e^{(3)}(0)\cdot t_a^3 + o\left(t_a^4\right)\right). \tag{A3}$$

In this study, the quadratic coefficient of motion error is estimated using complex correlation, i.e., the phase variation in the image domain. The phase variation caused by high-order terms should not exceed $\pi/4$ [50]. When the inequality in (A4) is satisfied, the influence of high-order terms is negligible

$$\left|-\frac{4\pi}{\lambda}\cdot\left(\frac{1}{6}\cdot R_e^{(3)}(0)\cdot t_a^3 + o\left(t_a^4\right)\right)\right| \le \frac{\pi}{4}. \tag{A4}$$

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When $(1/6)\cdot R_e^{(3)}(0)\cdot t_a^3 \gg o(t_a^4)$, the following boundary condition can be obtained:

$$\frac{4}{\lambda}\cdot\frac{|R_e^{(3)}(0)|}{6}\cdot t_a^3 \le \frac{1}{4}. \tag{A5}$$

Appendix B

The expressions from A_e to F_e are presented as follows:

$$\begin{aligned}
A_e &= (X(0)+\Delta X(0)-x_0)^2+(Y(0)+\Delta Y(0)-y_0)^2\\
&\quad+(H+\Delta Z(0))^2\\
B_e &= (X(0)-x_0)^2+(Y(0)-y_0)^2+H^2\\
C_e &= (X(0)+\Delta X(0)-x_0)\cdot\left(X'(0)+\Delta X'(0)\right)\\
&\quad+(Y(0)+\Delta Y(0)-y_0)\cdot\left(Y'(0)+\Delta Y'(0)\right)\\
&\quad+(H+\Delta Z(0))\cdot\Delta Z'(0)\\
D_e &= -(X(0)-x_0)\cdot X'(0)-(Y(0)-y_0)\cdot Y'(0)\\
E_e &= \left(X'(0)+\Delta X'(0)\right)^2+(X(0)+\Delta X(0)-x_0)\\
&\quad\cdot\left(X''(0)+\Delta X''(0)\right)\\
&\quad+\left(Y'(0)+\Delta Y'(0)\right)^2+(Y(0)+\Delta Y(0)-y_0)\\
&\quad\cdot\left(Y''(0)+\Delta Y''(0)\right)\\
&\quad+\left(\Delta Z'(0)\right)^2+(H+\Delta Z(0))\cdot\Delta Z''(0)\\
F_e &= -\left(X'(0)\right)^2-(X(0)-x_0)\cdot X''(0)-\left(Y'(0)\right)^2\\
&\quad-(Y(0)-y_0)\cdot Y''(0).
\end{aligned} \tag{B1}$$

Appendix C

Assume that the echo data of multiframe video SAR contain T azimuth pulses, which are uniformly divided into M subapertures. Thus, the number of pulses in each subaperture is $n_s = T/M$. Let the total azimuth duration corresponding to multiple frames be T_a and the pulse interval be Δt_a, where $\Delta t_a = T_a/T$.

Denote η_1 and η_2 as the initial and final pulse sequences of the mth subaperture, which can be expressed as

$$\begin{aligned}
\eta_1 &= (m-1)\cdot n_s+1\\
\eta_2 &= m\cdot n_s.
\end{aligned} \tag{C1}$$

Let ρ_{m-1} and ρ_m be the partition boundary of the mth subaperture in the azimuth time, which can be expressed as

$$\begin{aligned}
\rho_{m-1} &= t_a(\eta_1) = (\eta_1-1)\cdot\Delta t_a\\
\rho_m &= t_a(\eta_2) = (\eta_2-1)\cdot\Delta t_a.
\end{aligned} \tag{C2}$$

Based on these definitions, the set of pulses used in the mth subaperture satisfies the following relationship:

$$\rho_{m-1} < t_a(\eta) < \rho_m,\ \eta\in[\eta_1,\eta_2] \tag{C3}$$

where $t_a(\eta)$ denotes the azimuth time corresponding to the ηth pulse in the mth subaperture.

The constant, linear, and quadratic weights of the mth subaperture are defined according to the azimuth time of the pulses as follows:

$$\begin{aligned}
a_m &= \left(\frac{t_a(\eta)-\rho_{m-1}}{\rho_m-\rho_{m-1}}\right)^0\\
b_m &= \left(\frac{t_a(\eta)-\rho_{m-1}}{\rho_m-\rho_{m-1}}\right)^1\\
c_m &= \left(\frac{t_a(\eta)-\rho_{m-1}}{\rho_m-\rho_{m-1}}\right)^2,\ \eta\in[\eta_1,\eta_2].
\end{aligned} \tag{C4}$$

It is evident that the expressions for a_m, b_m, and c_m differ only in the exponents. Taking b_m as an example, the following presents the detailed expansion and simplification:

$$\begin{aligned}
b_m &= \frac{(\eta-1)\cdot\Delta t_a-(\eta_1-1)\cdot\Delta t_a}{(\eta_2-1)\cdot\Delta t_a-(\eta_1-1)\cdot\Delta t_a}\\
&= \frac{\eta-\eta_1}{\eta_2-\eta_1}=\frac{n_{\mathrm{re}}-1}{n_s-1},\ n_{\mathrm{re}}\in[1,n_s]
\end{aligned} \tag{C5}$$

where n_{re} represents the relative value of the azimuth pulse sequence in the subaperture. Similarly, the expressions for a_m and c_m can be simplified using the same approach, and the resulting forms are given as follows:

$$\begin{aligned}
a &= \left(\frac{n_{\mathrm{re}}-1}{n_s-1}\right)^0,\ a\in[0,1],\ n_{\mathrm{re}}\in[1,n_s]\\
b &= \left(\frac{n_{\mathrm{re}}-1}{n_s-1}\right)^1,\ b\in[0,1],\ n_{\mathrm{re}}\in[1,n_s]\\
c &= \left(\frac{n_{\mathrm{re}}-1}{n_s-1}\right)^2,\ c\in[0,1],\ n_{\mathrm{re}}\in[1,n_s].
\end{aligned} \tag{C6}$$

Appendix D

Taking the $\hat{m}$th subfunction segment of the Kth frame as an example for detailed derivation. First, each subfunction contains n_s discrete points, and the first point $(x_o^{\hat{m}}, y_o^{\hat{m}})$, the middle point $(x_h^{\hat{m}}, y_h^{\hat{m}})$, and the last point $(x_e^{\hat{m}}, y_e^{\hat{m}})$ are selected from the $y_o^{\hat{m}} = W_{2\hat{m}-1}$th subaperture. According to the definition of the fitting factor above, we can get $y_o^{\hat{m}} = W_{2\hat{m}-1}$, $y_h^{\hat{m}} = W_{2\hat{m}}$, and $y_e^{\hat{m}} = W_{2\hat{m}+1}$.

Next, to facilitate subsequent simplification, the independent variable of the fit function is normalized as $\vartheta = (x_\gamma^{\hat{m}}-x_o^{\hat{m}})\Big/(x_e^{\hat{m}}-x_o^{\hat{m}}),\ \gamma\in[1,n_s]$, where $x_\gamma^{\hat{m}}$ represents any discrete point in the $\hat{m}$th subfunction. Following the above definition, the normalized variable $\vartheta\in[0,1]$, and the three selected points are rewritten as $(0, W_{2\hat{m}-1})$, $(1/2, W_{2\hat{m}})$, and $(1, W_{2\hat{m}+1})$.

Finally, the fitting function is equivalent to a quadratic interpolation through the selected points. Using the Lagrange interpolation method, a quadratic polynomial $\tilde{fit}_{Ks} = C(\hat{m},K)\cdot\vartheta^2+B(\hat{m},K)\cdot\vartheta+A(\hat{m},K)\cdot 1$ is constructed to fit the $\hat{m}$th subfunction of the Kth frame. The fitting coefficients A_k, B_k, and C_k are determined based on the three selected points.

For the quadratic Lagrange interpolation, it is necessary to construct the corresponding basis functions $l_0(\vartheta), l_1(\vartheta)$, and $l_2(\vartheta)$ according to the three selected points

$$\begin{aligned}
l_0(\vartheta) &= \frac{(\vartheta-1/2)\cdot(\vartheta-1)}{(0-1/2)\cdot(0-1)} = 2\cdot(\vartheta-1/2)\cdot(\vartheta-1)\\
l_1(\vartheta) &= \frac{(\vartheta-0)\cdot(\vartheta-1)}{(1/2-0)\cdot(1/2-1)} = -4\cdot\vartheta\cdot(\vartheta-1)\\
l_2(\vartheta) &= \frac{(\vartheta-0)\cdot\left(\vartheta-\frac{1}{2}\right)}{(1-0)\cdot\left(1-\frac{1}{2}\right)} = 2\cdot\vartheta\cdot\left(\vartheta-\frac{1}{2}\right).
\end{aligned} \tag{D1}$$

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Based on the definition of the Lagrange interpolation polynomial $L(t) = \sum_{i=0}^{2} y_i \cdot l_i(t)$, the final fitting expression can be obtained as

$$\begin{aligned}
&\tilde{fit}_{Ks} \\
&= W_{2k-1} \cdot l_0(\vartheta) + W_{2k} \cdot l_1(\vartheta) + W_{2k+1} \cdot l_2(\vartheta) \\
&= W_{2k-1} \cdot \left(2\vartheta^2 - 3\vartheta + 1\right) - W_{2k} \cdot \left(4\vartheta^2 - 4\vartheta\right) + W_{2k+1} \\
&\quad \cdot \left(2\vartheta^2 - \vartheta\right) \\
&= (2W_{2k-1} + 2W_{2k+1} - 4W_{2k}) \cdot \vartheta^2 + (4W_{2k} - 3W_{2k-1} - W_{2k+1}) \\
&\quad \cdot \vartheta + (W_{2k-1}). \qquad \text{(D2)}
\end{aligned}$$

By comparison, the fitting coefficients $A(\hat{m}, K)$, $B(\hat{m}, K)$, and $C(\hat{m}, K)$ of the $\hat{m}$th subfunction can be expressed as follows:

$$\begin{aligned}
A\left(\hat{m}, K\right) &= W_{2\hat{m}-1} \\
B\left(\hat{m}, K\right) &= 4W_{2\hat{m}} - 3W_{2\hat{m}-1} - W_{2\hat{m}+1} \\
C\left(\hat{m}, K\right) &= 2W_{2\hat{m}-1} + 2W_{2\hat{m}+1} - 4W_{2\hat{m}}. \qquad \text{(D3)}
\end{aligned}$$

REFERENCES

[1] J. Ding, K. Zhang, X. Huang, and Z. Xu, "High frame-rate imaging using swarm of UAV-borne radars," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/TGRS.2024.3362630.

[2] H. B. Wallace, "Development of a video SAR for FMV through clouds," *Proc. SPIE*, vol. 9479, pp. 64–65, May 2015.

[3] A. Gao, B. Sun, J. Li, and C. Li, "A parameter-adjusting autoregistration imaging algorithm for video synthetic aperture radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2021.3115246.

[4] B. Zhao, Y. Han, H. Wang, L. Tang, X. Liu, and T. Wang, "Robust shadow tracking for video SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 821–825, May 2021, doi: 10.1109/LGRS.2020.2988165.

[5] R. Linnehan, J. Miller, E. Bishop, and V. Horndt, "An autofocus technique for video-SAR," *Proc. SPIE*, vol. 8746, pp. 56–65, May 2013.

[6] J. Miller, E. Bishop, and A. Doerry, "An application of backprojection for video SAR image formation exploiting a subaperature circular shift register," *Proc. SPIE*, vol. 8746, pp. 66–79, May 2013.

[7] J. Miller, E. Bishop, and A. Doerry, "Applying stereo SAR to remove height-dependent layover effects from video SAR imagery," *Proc. SPIE*, vol. 9093, pp. 99–108, Jun. 2014.

[8] W. G. Carrara, R. S. Goodman, and R. M. Majewski, *Spotlight Synthetic Aperture Radar: Signal Processing Algorithms*. Norwood, MA, USA: Artech House, 1995.

[9] C. Yang, F. Zhao, Y. Deng, K. Liu, F. Zhao, and W. Wang, "Extended polar format algorithm and video-SAR image generation scheme for very high-resolution curvilinear spotlight SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3310262.

[10] R. Linnehan, J. Miller, and A. Asadi, "Map-drift autofocus and scene stabilization for video-SAR," in *Proc. IEEE Radar Conf. (RadarConf)*, Oklahoma City, OK, USA, Apr. 2018, pp. 1401–1405, doi: 10.1109/RADAR.2018.8378769.

[11] O. O. Bezvesilniy, I. M. Gorovyi, and D. M. Vavriv, "Estimation of phase errors in SAR data by local-quadratic map-drift autofocus," in *Proc. 13th Int. Radar Symp.*, Warsaw, Poland, May 2012, pp. 376–381, doi: 10.1109/IRS.2012.6233350.

[12] Y. Xiang, F. Wang, and H. You, "OS-SIFT: A robust SIFT-like algorithm for high-resolution optical-to-SAR image registration in suburban areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3078–3090, Jun. 2018, doi: 10.1109/TGRS.2018.2790483.

[13] P. Etezadifar and H. Farsi, "A new sample consensus based on sparse coding for improved matching of SIFT features on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5254–5263, Aug. 2020, doi: 10.1109/TGRS.2019.2959606.

[14] F. Dellinger, J. Delon, Y. Gousseau, J. Michel, and F. Tupin, "SAR-SIFT: A SIFT-like algorithm for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 453–466, Jan. 2015, doi: 10.1109/TGRS.2014.

[15] M. Wang, J. Zhang, K. Deng, and F. Hua, "Combining optimized SAR-SIFT features and RD model for multisource SAR image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2021.3074630.

[16] B. Zitová and J. Flusser, "Image registration methods: A survey," *Image Vis. Comput.*, vol. 21, no. 11, pp. 977–1000, Oct. 2003, doi: 10.1016/S0262-8856(03)00137-9.

[17] M. Gong, S. Zhao, L. Jiao, D. Tian, and S. Wang, "A novel coarse-to-fine scheme for automatic image registration based on SIFT and mutual information," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4328–4338, Jul. 2014, doi: 10.1109/TGRS.2013.2281391.

[18] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004, doi: 10.1023/b:visi.0000029664.99615.94.

[19] Y. Ke and R. Sukthankar, "PCA-SIFT: A more distinctive representation for local image descriptors," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit., CVPR.*, vol. 2, Washington, DC, USA, Jul. 2004, pp. 506–513, doi: 10.1109/CVPR.2004.1315206.

[20] H. Bay, T. Tuytelaars, and L. Van Gool, "SURF: Speeded up robust features," in *Computer Vision—ECCV 2006*, A. Leonardis, H. Bischof, and A. Pinz, Eds., Berlin, Germany: Springer, 2006, pp. 404–417.

[21] P. Moreno, A. Bernardino, and J. Santos-Victor, "Improving the SIFT descriptor with smooth derivative filters," *Pattern Recognit. Lett.*, vol. 30, no. 1, pp. 18–26, Jan. 2009, doi: 10.1016/j.patrec.2008.08.012.

[22] S. Wang, H. You, and K. Fu, "BFSIFT: A novel method to find feature matches for SAR image registration," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 4, pp. 649–653, Jul. 2012, doi: 10.1109/LGRS.2011.2177437.

[23] P. Schwind, S. Suri, P. Reinartz, and A. Siebert, "Applicability of the SIFT operator to geometric SAR image registration," *Int. J. Remote Sens.*, vol. 31, no. 8, pp. 1959–1980, Apr. 2010, doi: 10.1080/01431160902927622.

[24] A. Sedaghat, M. Mokhtarzade, and H. Ebadi, "Uniform robust scale-invariant feature matching for optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4516–4527, Nov. 2011, doi: 10.1109/TGRS.2011.2144607.

[25] S. Paul and U. C. Pati, "SAR image registration using an improved SAR-SIFT algorithm and delaunay-triangulation-based local matching," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 2958–2966, Aug. 2019, doi: 10.1109/JSTARS.2019.2918211.

[26] A. Sedaghat and H. Ebadi, "Remote sensing image matching based on adaptive binning SIFT descriptor," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5283–5293, Oct. 2015, doi: 10.1109/TGRS.2015.

[27] M. Xing, P. Ma, Y. Lou, G. Sun, and H. Lin, "Review of fast back projection algorithms in synthetic aperture radar," *J. Radars*, vol. 13, no. 1, pp. 1–22, Feb. 2024, doi: 10.12000/JR23183.

[28] A. F. Yegulalp, "Fast backprojection algorithm for synthetic aperture radar," in *Proc. IEEE Radar Conf. Radar Into Next Millennium*, Apr. 1999, pp. 60–65, doi: 10.1109/NRC.1999.767270.

[29] X. Song and W. Yu, "Processing video-SAR data with the fast back-projection method," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 52, no. 6, pp. 2838–2848, Dec. 2016, doi: 10.1109/TAES.2016.150581.

[30] R. W. Hawley and W. L. Garber, "Aperture weighting technique for video synthetic aperture radar," *Proc. SPIE*, vol. 8051, pp. 67–73, May 2011.

[31] D. E. Wahl, D. A. Yocky, and C. V. Jakowatz Jr., "An implementation of a fast backprojection image formation algorithm for spotlight-mode SAR," *Proc. SPIE*, vol. 6970, pp. 95–105, Apr. 2008.

[32] R. L. Moses and J. N. Ash, "An autoregressive formulation for SAR backprojection imaging," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 47, no. 4, pp. 2860–2873, Oct. 2011, doi: 10.1109/TAES.2011.6034669.

[33] G. Xu, B. Zhang, H. Yu, J. Chen, M. Xing, and W. Hong, "Sparse synthetic aperture radar imaging from compressed sensing and machine learning: Theories, applications, and trends," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 4, pp. 32–69, Dec. 2022, doi: 10.1109/MGRS.2022.3218801.

[34] G. Xu, B. Zhang, J. Chen, F. Wu, J. Sheng, and W. Hong, "Sparse inverse synthetic aperture radar imaging using structured low-rank method," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2021.3118083.

[35] W. Pu, X. Wang, J. Wu, Y. Huang, and J. Yang, "Video SAR imaging based on low-rank tensor recovery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 188–202, Jan. 2021, doi: 10.1109/TNNLS.2020.

[36] N. Cao et al., "Estimation of residual motion errors in airborne SAR interferometry based on time-domain backprojection and multi-squint techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2397–2407, Apr. 2018, doi: 10.1109/TGRS.2017.2779852.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:37:40 UTC from IEEE Xplore. Restrictions apply.

[37] J. Torgrimsson, P. Dammert, H. Hellsten, and L. M. H. Ulander, "Factorized geometrical autofocus for synthetic aperture radar processing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 10, pp. 6674–6687, Oct. 2014, doi: 10.1109/TGRS.2014.2300347.

[38] J. Torgrimsson, P. Dammert, H. Hellsten, and L. M. H. Ulander, "An efficient solution to the factorized geometrical autofocus problem," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4732–4748, Aug. 2016, doi: 10.1109/TGRS.2016.2550663.

[39] J. Torgrimsson, P. Dammert, H. Hellsten, and L. M. H. Ulander, "Factorized geometrical autofocus for UWB UHF-band SAR with a GPS-supported linear track model," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 4, pp. 3147–3161, Aug. 2022, doi: 10.1109/TAES.2022.3149235.

[40] L. Ran, Z. Liu, L. Zhang, T. Li, and R. Xie, "An autofocus algorithm for estimating residual trajectory deviations in synthetic aperture radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3408–3425, Jun. 2017, doi: 10.1109/TGRS.2017.2670785.

[41] M. Shil, H. Rakshit, and H. Ullah, "An adjustable window function to design an FIR filter," in *Proc. IEEE Int. Conf. Imag., Vis. Pattern Recognit. (icIVPR)*, Dhaka, Bangladesh, Feb. 2017, pp. 1–5, doi: 10.1109/ICIVPR.2017.7890865.

[42] P. Putranto et al., "Performance comparison of blackman, bartlett, hanning, and Kaiser window for radar digital signal processing," in *Proc. 4th Int. Conf. Inf. Technol., Inf. Syst. Electr. Eng. (ICITISEE)*, Yogyakarta, Indonesia, Nov. 2019, pp. 391–394, doi: 10.1109/ICITISEE48480.2019.9003791.

[43] R. Zhu, J. Zhou, Q. Fu, and G. Jiang, "Spatially variant apodization for grating and sidelobe suppression in near-range MIMO array imaging," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 11, pp. 4662–4671, Nov. 2020, doi: 10.1109/TMTT.2020.2995706.

[44] L. Zhang, H.-L. Li, Z.-J. Qiao, M.-D. Xing, and Z. Bao, "Integrating autofocus techniques with fast factorized back-projection for high-resolution spotlight SAR imaging," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 6, pp. 1394–1398, Nov. 2013, doi: 10.1109/LGRS.2013.2258886.

[45] Z. Xu, J. Ding, G. Chen, G. Zhao, and Y. Zhang, "An approach to high frame rate radar imaging through electronically displaced-phase-center antenna," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5602–5611, Aug. 2020, doi: 10.1109/TGRS.2020.2967639.

[46] H. Chang and W. Chan, "Automatic registration of remote sensing images based on revised SIFT with trilateral computation and homogeneity enforcement," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7635–7650, Sep. 2021, doi: 10.1109/TGRS.2021.3052926.

[47] M. A. Fischler and R. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Commun. ACM*, vol. 24, no. 6, pp. 381–395, 1981.

[48] Y. Chi and C. Liu, "An improved SIFT algorithm for SAR image registration," *J. Univ. Chin. Acad. Sci.*, vol. 36, no. 2, pp. 259–266, Mar. 2019, doi: 10.7523/j.issn.2095-6134.2019.02.014.

[49] E. Karami, S. Prasad, and M. Shehata, "Image matching using SIFT, SURF, BRIEF and ORB: Performance comparison for distorted images," 2017, *arXiv:1710.02726*.

[50] S. Shao, H. Liu, L. Zhang, P. Wang, and J. Wei, "Ultrawideband ISAR imaging of maneuvering targets with joint high-order motion compensation and azimuth scaling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5214621, doi: 10.1109/TGRS.2021.3113007.



Shuo Wang received the B.S. degree from Beijing University of Civil Engineering and Architecture, Beijing, China, in 2019, and the master's degree from North China University of Technology, Beijing, in 2023, where he is currently pursuing the Ph.D. degree with the School of Electrical and Control Engineering.

His research interests include synthetic aperture radar (SAR) imaging and SAR moving target detection.



Guanyong Wang was born in Tianjin, China, in 1989. He received the Ph.D. degree from the Second Academy of China Aerospace Science and Industry Corporation (CASIC), Beijing, China, in 2018.

He is currently working as a Senior Engineer with the School of Artificial Intelligence and Computer Science, North China University of Technology, Beijing. His research interests include synthetic aperture radar (SAR) imaging and target detection.



Yanping Wang received the bachelor's and master's degrees from the Beijing Institute of Technology, Beijing, China, in 1998 and 2001, respectively, and the doctorate degree from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2004.

He is currently a Professor and a Researcher with North China University of Technology, Beijing. His research interests include synthetic aperture radar (SAR) imaging technology and ground-based SAR system.



Rui Zhou was born in Shandong, China, in 1987. He received the M.S. degree from Beihang University (BUAA), Beijing, China, in 2012.

He is currently working as a Senior Engineer with Beijing Institute of Radio Measurement, Beijing. His research interests include the design and signal processing of synthetic aperture radar (SAR) systems.



Meng Zhao was born in Hebei, China, in 1999. She received the B.S. degree in communication engineering from North China Institute of Science and Technology, Langfang, China, in 2021. She is currently pursuing the M.S. degree with the School of Artificial Intelligence and Computer Science, North China University of Technology, Beijing, China.

Her main research interests include synthetic aperture radar (SAR) target detection.



Yuxiao Wang was born in Shandong, China, in 2003. He received the B.S. degree in communication engineering from Beijing Jiaotong University, Beijing, China, in 2024. He is currently pursuing the M.S. degree with the School of Information and Electronics, Beijing Institute of Technology, Beijing.

His research interests include space-surface bistatic synthetic aperture radar and synthetic aperture radar (SAR) imaging.

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